Hybrid Hidden Markov Model for Modeling Equity Excess Growth Rate Dynamics: A Discrete-State Approach with a Jump-Duration Mechanism

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Abstract

Synthetic equity return paths support stress testing and portfolio validation and serve as training data for machine-learning models, but parametric, neural, and semi-Markov generators rarely reproduce heavy tails, regime persistence, and volatility clustering together. We developed a hybrid hidden Markov model with a jump-duration mechanism (HMM-WJ): excess growth rates are assigned to Laplace-quantile states, each with a location-scale Student-t emission, and Poisson-distributed jump episodes lengthen tail-state dwell times. We fitted HMM-WJ and a no-jump variant (HMM-NJ) to a decade of SPY history and evaluated both, in and out of sample, against a generalized autoregressive conditional heteroskedasticity model (GARCH(1,1)) and five parametric, semi-Markov, and neural baselines. Relative to HMM-NJ, HMM-WJ retained high distributional fit and reduced absolute-growth autocorrelation error from 0.059 to 0.052, though GARCH still had the lowest error at 0.031. To extend the model to multiple assets, we used a single-index model to compose paths for 423 non-market tickers in a 424-asset US equity and exchange-traded-fund universe. A closed-form scale correction removed the resulting variance inflation, recovering each ticker's calibrated factor loading while retaining heavy-tailed returns. Residual models fit directly to each ticker still matched the distribution's center more closely. Independent per-asset residuals can overstate diversification gains under daily rebalancing. A location-shift correction reduces this bias but does not restore the missing cross-asset dependence. Code and per-asset fitted marginals are released with the paper.

Keywords: hidden Markov model; jump-duration mechanism; single-index model; synthetic financial data; variance correction; Value-at-Risk coverage check; volatility clustering; heavy-tailed marginals.

1 Introduction

Synthetic financial data often fail to reproduce the statistical properties of real returns. These paths are used to stress-test risk models against unobserved market scenarios, validate portfolio optimization

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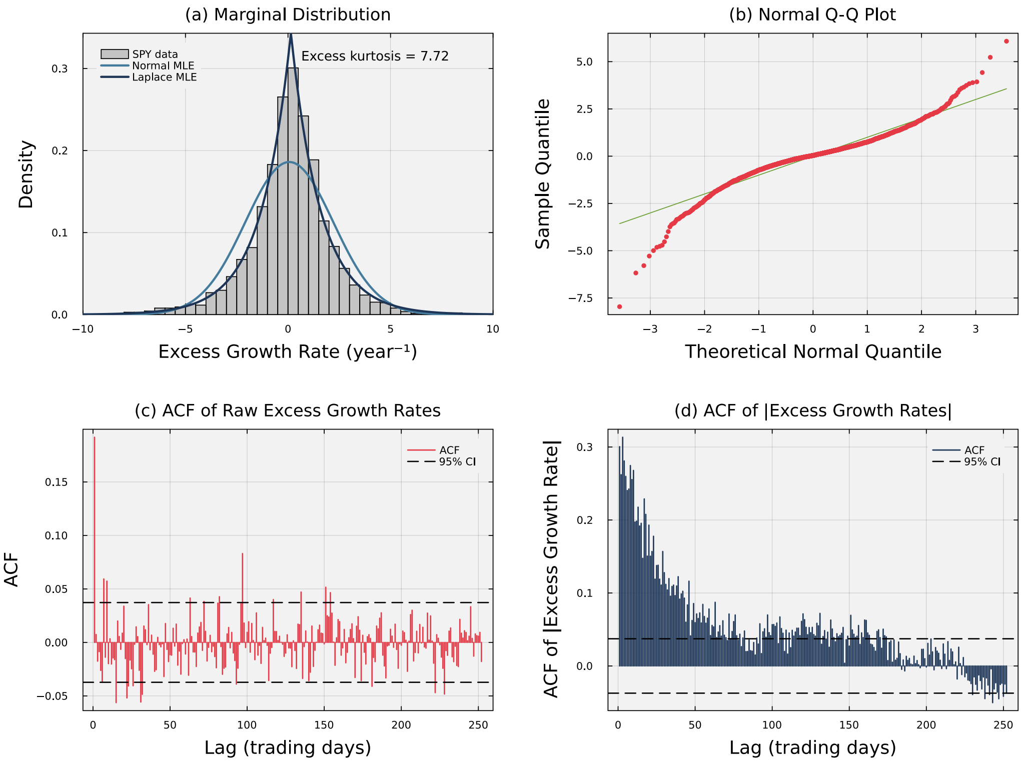

Figure 1: **Empirical stylized facts of SPY daily excess growth rates** (2014–2024)**.** Panel (a) shows the leptokurtic marginal density; the Laplace maximum-likelihood fit tracks the empirical peak and tails while the Gaussian fit does not. Panel (b) confirms heavy tails through a normal quantile-quantile plot, with departure from the diagonal at both extremes. Panel (c) shows the autocorrelation function of raw excess growth rates, which stays near zero across all lags and is consistent with the efficient markets hypothesis. Panel (d) shows the autocorrelation function of absolute excess growth rates, which decays slowly and signals the persistent volatility clustering that the jump-duration extension is designed to reproduce.

algorithms, and Monte Carlo backtest allocators across many independent scenarios [1, 2]. Empirical equity excess growth rates exhibit three statistical regularities a generative model must reproduce: a heavy-tailed marginal distribution, weak autocorrelation of raw growth rates, and persistent volatility clustering, in which large changes are followed by large changes and small changes by small changes [3–5]. All three patterns are visible in SPY daily excess growth rates from 2014–2024 (Fig. 1). These regularities, called *stylized facts*, are the benchmark for synthetic financial data.

Existing generative approaches typically reproduce some of these regularities but not others. The generalized autoregressive conditional heteroskedasticity (GARCH) family [6, 7] captures volatility clustering but needs a fat-tailed innovation to reproduce heavy tails; stochastic-volatility and jump-diffusion models [8–10] enrich tail behavior but do not persist after a jump; deep generative models [11–13] learn complex marginals but routinely miss the absolute-return autocorrelation that defines volatility clustering; and standard hidden Markov models (HMMs) [14] offer a regime-switching structure but revert to central regimes too quickly to dwell in tail states long enough to reproduce the empirical clustering profile [15, 16]. We therefore sought a model that retained distributional fit while improving volatility persistence and remaining practical for large asset universes.

In this study, we developed a hybrid hidden Markov model with a jump-duration mechanism (HMM-WJ) that pairs a Laplace-quantile state partition with a Student-t emission and a Poisson-distributed jump-duration override. The Laplace partition lets the transition matrix be estimated by direct frequentist counting rather than the iterative Baum-Welch algorithm [17, 18], which scales to a 424-asset universe. We fitted the model to a decade of SPY history, held out 2025 for out-of-sample evaluation, and scored the synthetic paths against seven baseline generators on distributional and temporal fidelity. Relative to a no-jump variant, HMM-WJ retained high distributional fit while improving volatility-clustering fidelity, though GARCH remained the strongest baseline on that dimension; HMM-WJ is therefore a tradeoff between marginal and temporal fit rather than a uniformly better model. Extending the model to multi-asset applications required correcting a known limitation of single-index composition, in which drawing each asset's residual from a per-asset generator inflates the composed marginal's variance quadratically in the factor loading; we derived a closed-form scale correction that restores each ticker's target variance after factor composition, recovering each ticker's calibrated factor loading while retaining heavy-tailed returns. Independent per-asset residuals also exaggerate the daily cross-sectional spread among assets, so daily-rebalanced portfolios built from these paths can overstate diversification gains; we report this bias and a location-shift adjustment set to a long-run prior, but the adjustment does not restore the missing cross-asset dependence.

2 Related Work

Approaches to reproducing the stylized facts of equity markets divide into bottom-up agent-based models [19–24], which derive return dynamics from simulated interactions of heterogeneous agents but are demanding to calibrate and expensive to simulate at scale, and top-down models, which fit statistical structures directly to observed growth-rate series. Top-down generators trace to Bachelier's diffusion model, later formalized as geometric Brownian motion and used by Black-Scholes-Merton for option pricing [25–28]; these diffusion baselines are tractable but fail on the empirical regularities that drive risk and pricing. Mandelbrot [3] showed returns have heavier tails than the Gaussian admits; Fama [29] reviewed evidence that raw returns have near-zero autocorrelation; and Schwert [30] documented persistence in monthly absolute residuals, later codified by Cont [4] as volatility clustering. The autoregressive conditional heteroskedasticity (ARCH) and generalized ARCH (GARCH) models of Engle and Bollerslev [6, 7] reproduced the clustering but left marginals light-tailed unless the noise term was itself heavy-tailed, and neither models discrete regimes or sudden jumps. Stochastic-volatility models [31, 8] make volatility a latent process to capture clustering, but estimating that process requires filtering or Markov chain Monte Carlo (MCMC) methods [32]; jump-diffusion models [9, 10] inject Poisson events into the price path, producing heavy tails by construction, but volatility does not persist after the jump.

Hidden Markov models embed regime and jump behavior directly in a latent discrete chain, following Hamilton's [33] two-state regime-switching formulation, later applied to volatility forecasting [34] and equity trading [35]. Ryden et al. [15] found that Gaussian-emission HMMs matched heavy tails and the absence of return autocorrelation but failed on volatility clustering; Bulla and Bulla [16] recovered the clustering with hidden semi-Markov models (HSMMs) that replaced the geometric dwell-time distribution of every state with a parametric (negative-binomial) alternative. Both rely on the Baum-Welch expectation-maximization (EM) algorithm [17], which is iterative and sensitive to initialization. Deep generative alternatives learn the marginal directly but routinely miss the volatility-clustering signature: Takahashi et al. [11] found that generative adversarial network (GAN) architectures produced return sequences that resembled financial data yet failed to reproduce the absolute-return autocorrelation function, and Kwon and Lee [13] confirmed the same on TimeGAN [12]. Recent surveys [1, 2, 5] agree that stylized-facts reproduction remains the primary quality criterion. This model also uses a latent-state chain but follows a different design principle than the HSMM: rather than replace every state's dwell-time distribution, it keeps the empirically counted transition matrix for the central states and adds an event-driven override that forces the chain into the tail states for a Poisson-distributed duration on a subset of time steps.

Multi-asset composition adds cross-asset dependence on top of a univariate generator. Mixture hidden Markov models [36] share latent state across assets but are costly to estimate in high dimensions; copula approaches [37–39] separate marginals from cross-asset dependence but require choosing a copula family and, in their standard static form, treat successive time slices as independent. Factor models instead reduce a firm's growth rate to one or more shared factors [40, 41]; the single-index model (SIM) of Sharpe [42] is the simplest, decomposing each asset's growth rate into a market-loaded term plus an idiosyncratic residual, a linear, easy-to-interpret structure that pairs naturally with a single-factor market generator. Within SIM, the choice of residual specification trades off the per-asset generator's distributional structure against the composed variance; we derive a closed-form variance correction that recovers both.

Validation of synthetic growth-rate paths combines two families of tests: goodness-of-fit tests for distributional fidelity [43–45] and autocorrelation diagnostics for temporal structure [4]. These tests are necessary but not sufficient: a synthetic dataset can pass them yet still mislead when fed into the task it was generated for. Booth and Fama [46] described one such failure: a daily-rebalanced allocator on paths whose residuals are drawn independently across assets extracts a structural *diversification return* that is muted in real markets, where common sector and style factors correlate residuals across assets. We adopt the classical distributional metrics together with a Value-at-Risk coverage check scored by Kupiec's unconditional-coverage test, and address this diversification-return bias with a location-shift correction.

3 Methods

3.1 Hybrid hidden Markov marginal generator

The dataset was 424 United States-listed equities and exchange-traded funds spanning ten years (2014–2024). To validate the hybrid hidden Markov model on a single representative asset, we focused the main analysis on the SPDR S&P 500 ETF (SPY), reserving the $2,766$ observations from January 3, 2014 to December 31, 2024 for in-sample fitting and the subsequent 249 trading days of 2025 for out-of-sample testing. The data consisted of daily open, high, low, and close prices and volume-weighted average price values. We computed the excess growth rate $G_{i,j}$ for ticker i between trading days $j-1$ and j from the equity price series $P_{i,j}$ as:

$$G_{i,j} \;\equiv\; \left(\frac{1}{\Delta t}\right) \ln\!\left(\frac{P_{i,j}}{P_{i,j-1}}\right) \;-\; r_f, \tag{1}$$

where $\Delta t = 1/252$ is the fraction of a trading year spanned by one trading day, and r_f is a constant continuously compounded risk-free rate derived from Treasury STRIPS (Separate Trading of Registered Interest and Principal of Securities) bond yields. The excess growth rate has units of year^{-1} and is time-additive under continuous compounding.

We defined the hybrid hidden Markov model with a jump-duration mechanism (HMM-WJ) as the tuple $\mathcal{M} = (\mathcal{S}, \mathcal{O}, \mathbf{T}, \mathbf{E}, \bar{\pi})$ [14] (Figure 2). The hidden state $S_t \in \mathcal{S} = \{1, \ldots, N\}$ represents a market regime, and the observation at step t is the excess growth rate G_t of Eq. (1), drawn from the continuous observation space $\mathcal{O} \subset \mathbb{R}$ (we suppress the ticker index i when only one asset is in play). The transition matrix $\mathbf{T}$ governs regime-to-regime transitions, $\mathbf{E}$ is the state-conditional emission model, and $\bar{\pi}$ is the stationary distribution.

States were defined by quantile boundaries of a fitted Laplace cumulative distribution: $Q_k = F_L^{-1}(k/N;\, \mu_L, b_L)$, where $F_L^{-1}(\cdot;\, \mu_L, b_L)$ is the inverse cumulative distribution function of the Laplace distribution with location μ_L and scale b_L (both estimated from the in-sample growth-rate history by maximum likelihood), with the endpoints fixed at $Q_0 = -\infty$ and $Q_N = +\infty$. An observation was assigned to state k when $Q_{k-1} < G_t \le Q_k$. We chose the Laplace distribution for its sharper peak, which better matches the concentration of small price movements in the empirical marginal density (Fig. 1(a)) [47, 48], and for its closed-form quantile function, which scales to the 424-ticker universe without iterative optimization [18].

The within-state emission followed a location-scale Student-t distribution:

$$G_t \mid S_t = k \;\sim\; \mu_k + \sigma_k \cdot t_\nu, \qquad \nu = 5, \tag{2}$$

where t_ν denotes a standard Student-t random variable with ν degrees of freedom, μ_k is the sample mean of the in-sample observations assigned to state k, and σ_k is their sample standard deviation; because a standard t_ν has variance $\nu/(\nu-2)$, σ_k is a scale rather than the emission's actual standard deviation, which is $\sigma_k\sqrt{5/3}$ at $\nu = 5$. We selected $\nu = 5$ from a sensitivity sweep ($\nu = \infty$ recovers the Normal baseline; explored range in Table 1) as the value that minimized the kurtosis gap without degrading distributional fidelity; the full comparison against a Gaussian-emission HSMM baseline of Bulla and Bulla [16] is given in Table S3 of Sec. S5.

With the state space and emission family fixed, we estimated the transition matrix $\mathbf{T}$ by counting observed state-transition frequencies directly, avoiding the initialization sensitivity of iterative expectation-maximization [18]. Direct counting, however, gives empirical transition matrices low probabilities for exiting central states into tail states, so models built this way lose the effect of volatility clustering too quickly [16]. We corrected this by adding two jump hyperparameters, ϵ representing

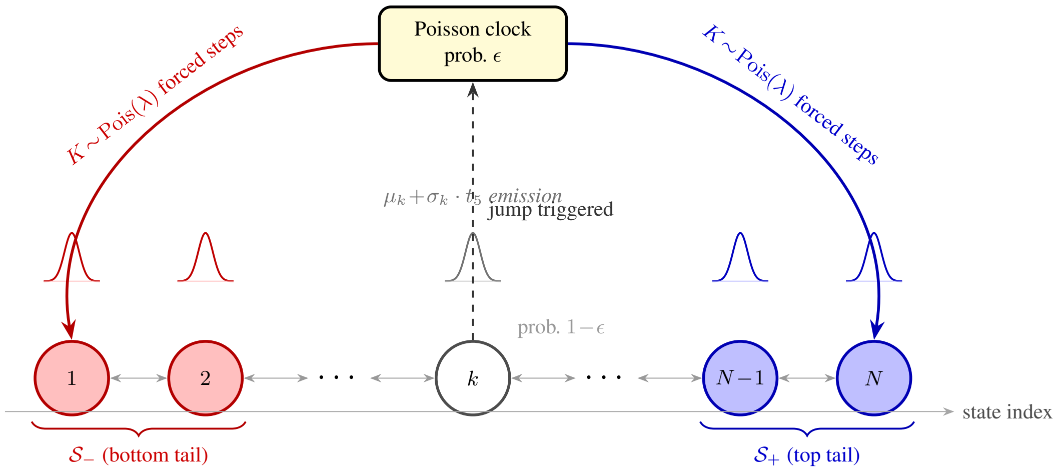

Figure 2: **Architecture of the hybrid hidden Markov model with a jump-duration mechanism (HMM-WJ).** At each step the chain transitions according to the empirical transition matrix $\mathbf{T}$ (probability $1-\epsilon$, thin bidirectional arrows) or enters a Poisson jump episode (probability ϵ, dashed upward arrow to the Poisson clock). During a jump episode each of the $K \sim \mathrm{Poisson}(\lambda)$ forced steps independently draws the bottom tail set $\mathcal{S}_-$ (red, lowest-valued states) with probability p_{neg} or the top tail set $\mathcal{S}_+$ (blue, highest-valued states) otherwise, before normal Markovian evolution resumes. Small bell curves above each state depict the state-conditional location-scale Student-t ($\nu = 5$) emission distribution. Tail sets are defined as the N_{tail} lowest- and highest-quantile states under the Laplace quantile partition.

the probability of a jump event and λ representing the mean jump-event duration, along with a tail-state selection rule. Tail states were defined as $\mathcal{S}_- = \{1, \ldots, N_{\text{tail}}\}$ and $\mathcal{S}_+ = \{N - N_{\text{tail}} + 1, \ldots, N\}$, where N_{tail} was user-configurable (value in Table 1). When a jump was triggered, the jump-episode length K was sampled from a Poisson distribution $K \sim \mathrm{Poisson}(\lambda)$ [49]; since the Poisson support includes zero, a draw of $K = 0$ forces no tail steps and the chain takes an ordinary transition, so λ sets the mean length of a triggered episode rather than a guaranteed minimum dwell. During the K forced steps, the standard Markovian transitions were overridden to target tail states: at each forced step the tail set was drawn as $\mathcal{S}_-$ with a user-configurable probability p_{neg} and as $\mathcal{S}_+$ otherwise (values in Table 1), a bias toward negative-tail states that reproduced gain/loss asymmetry (Figure 2). The empirical transition matrix exhibited a dominant near-diagonal band reflecting short-range regime persistence, with natural residence times of 1–2 steps in every state including the tails (Fig. S5 in Online Appendix S4). The stationary distribution $\bar{\pi}$ satisfies $\bar{\pi} = \bar{\pi}\mathbf{T}$ and was estimated by matrix powering ($\mathbf{T}^{50}$) [50, 51]; the initial state was drawn from $S_0 \sim \bar{\pi}$. Because $\bar{\pi}$ is the invariant distribution of $\mathbf{T}$ alone, it only approximates the invariant distribution of the jump-augmented process, whose state also includes jump status and remaining episode length; this introduces a small, unquantified deviation from exact stationarity. Algorithm 1 gives the full simulator: at each step the chain takes a standard transition with probability $1-\epsilon$ or, otherwise, enters a Poisson jump episode that forces K consecutive draws from a tail set before normal transitions resume, sampling an observation $G_t \sim \mathbf{E}_{S_t}$ at every step.

We calibrated the two jump hyperparameters, the jump probability ϵ and mean episode length λ, drawing on the jump-diffusion tradition of [9, 10] while adopting an HSMM-style state-duration parameterization [16] in place of the instantaneous-jump formulation of those classical models. As a discrete-time state model, HMM-WJ lengthens dwell time in tail states through the jump-duration override and includes no continuous-time diffusion component. Calibration used a multi-objective grid search that minimized the discrepancy between historical and simulated market signatures, targeting two stylized facts [4, 15] visible in the empirical SPY history of Fig. 1: volatility clustering (Fig. 1(d)), measured as the squared error between the observed and simulated autocorrelation function (ACF) of absolute excess growth rates [4] up to a lag of L trading days (value in Table 1), the range where most of the observed ACF variation occurs (post-fit evaluation in Fig. 3 extends to lag 252 for the visual comparison), and leptokurtosis (Fig. 1(a, b)), captured by a penalty term on the

Algorithm 1 HMM-WJ single asset marginal generator.

Require: Fitted parameters $(\mathbf{T}, \mathbf{E}, \bar{\pi}, \epsilon, \lambda, p_{\text{neg}}, N_{\text{tail}})$; horizon T; tail sets $\mathcal{S}_- = \{1, \ldots, N_{\text{tail}}\}$ and $\mathcal{S}_+ = \{N - N_{\text{tail}} + 1, \ldots, N\}$.
Ensure: State sequence $\{S_t\}_{t=0}^{T}$ and growth-rate path $\{G_t\}_{t=1}^{T}$.

Draw $S_0 \sim \bar{\pi}$; $k \leftarrow 0$ {k: remaining forced-step counter}
for $t = 1, \ldots, T$ **do**
  if $k > 0$ **then**
    $\mathcal{J} \leftarrow \mathcal{S}_-$ w.p. p_{neg}, else $\mathcal{S}_+$; $S_t \sim \text{Uniform}(\mathcal{J})$; $k \leftarrow k - 1$ {forced tail step; sign drawn per step}
  else if $\text{Uniform}(0, 1) > \epsilon$ **then**
    $S_t \sim \mathbf{T}_{S_{t-1},\cdot}$ {standard Markov transition}
  else
    $K \sim \text{Poisson}(\lambda)$ {trigger jump; K is the episode length}
    if $K = 0$ **then**
      $S_t \sim \mathbf{T}_{S_{t-1},\cdot}$ {$K = 0$: no forced steps, standard transition}
    else
      $\mathcal{J} \leftarrow \mathcal{S}_-$ w.p. p_{neg}, else $\mathcal{S}_+$; $S_t \sim \text{Uniform}(\mathcal{J})$; $k \leftarrow \min(K - 1, T - t)$ {first forced step; K steps total, sign per step}
    end if
  end if
  Draw $G_t \sim \mu_{S_t} + \sigma_{S_t}\, t_5$ {Eq. (2)}
end for

global kurtosis [3]:

$$J(\epsilon, \lambda) = \sum_{\tau=1}^{L} \left(\text{ACF}_{\text{obs}}(\tau) - \overline{\text{ACF}}_{\text{sim}}(\tau) \right)^2 + w_\kappa \left(\kappa_{\text{obs}} - \overline{\kappa}_{\text{sim}} \right)^2, \tag{3}$$

where $\text{ACF}_{\text{obs}}(\tau)$ and κ_{obs} are the empirical absolute-growth autocorrelation at lag τ and the global kurtosis, respectively (kurtosis denoted κ to avoid collision with the jump-duration variable K), and the simulated counterparts $\overline{\text{ACF}}_{\text{sim}}(\tau)$ and $\overline{\kappa}_{\text{sim}}$ were averaged over the jump-active paths among 200 independent synthetic paths of 2,766 trading days simulated per grid point: a path that fires no jump reduces to the no-jump generator and provides no information about (ϵ, λ), so the objective is conditioned on paths where at least one jump occurred [49]. We set the kurtosis-penalty weight w_κ to balance tail behavior against the temporal ACF term and explored (ϵ, λ) by brute-force grid search; the swept ranges, w_κ, L, and every other HMM-WJ and SIM composition hyperparameter are collected in Table 1. The resulting model produced a per-ticker marginal that reproduces heavy tails, regime persistence, and volatility clustering.

3.2 Multi-asset extension

The HMM-WJ model produces a per-ticker marginal, but does not capture cross-asset dependence that is required for multi-asset applications. To generalize synthetic path generation beyond a single asset, we composed the per-ticker marginals using the single-index model (SIM) of Sharpe [42], which decomposes each asset's excess growth rate as:

$$g_i(t) \;=\; \alpha_i \;+\; \beta_i\, g_m(t) \;+\; \varepsilon_i(t), \qquad t = 1, \ldots, T, \tag{4}$$

where $g_m \in \mathbb{R}^T$ is the market growth-rate path with sample variance $\sigma_m^2 = \text{Var}(g_m)$, the pair (α_i, β_i) is the SIM intercept and slope for asset i, and ε_i is the per-asset residual that the construction must specify (unrelated to the jump probability ϵ of the marginal generator). Two constraints govern the choice of ε_i: the composed g_i must recover the calibrated (α_i, β_i) under ordinary least squares (OLS), and it must inherit the per-asset marginal the HMM-WJ generator was validated to reproduce (heavy tails, regime structure, jumps).

A first option, *Gaussian-residual composition*, fits a Normal distribution to the OLS residuals of the asset's real history on g_m and samples $\varepsilon_i \sim N(0, \sigma_{\varepsilon,i}^2)$ with $\sigma_{\varepsilon,i}^2 = (1 - R_{i,\text{real}}^2)\,\sigma_i^2$, where $R_{i,\text{real}}^2$ is the OLS coefficient of determination from regressing the asset's history on g_m and σ_i^2 is the empirical variance of the asset's excess growth rates. This recovers (α_i, β_i) and pins $\text{Var}(g_i)$ to σ_i^2, but the resulting g_i lacks the heavy tails, regime structure, and jumps the per-asset HMM-WJ generator was built to reproduce. A separately fitted residual distribution, whether Gaussian, Laplace, Student-t, or empirical bootstrap, does not retain the fitted full-return marginal.

A second option uses the single-asset generator path $\tilde{g}_i$ (with sample variance $\sigma^2_{\text{gen},i}$) directly as the residual:

$$g_i = \alpha_i + \beta_i\, g_m + \tilde{g}_i \tag{5}$$

This approach, which we call *generator-residual composition*, captures the empirical per-asset marginal with variance $\sigma^2_{\text{gen},i} \approx \sigma_i^2$ calibrated to the asset, so its tails, regimes, and jumps are correct by construction. However, this approach also inflates the marginal variance. To see this, take the variance of both sides of Eq. (5):

$$\begin{aligned}
\mathrm{Var}(g_i) &= \mathrm{Var}(\alpha_i + \beta_i\, g_m + \tilde{g}_i) && \text{(naive composition)}\\
&= \mathrm{Var}(\beta_i\, g_m + \tilde{g}_i) && (\alpha_i \text{ is a constant})\\
&= \beta_i^2\ \mathrm{Var}(g_m) + 2\beta_i\ \mathrm{Cov}(g_m,\, \tilde{g}_i) + \mathrm{Var}(\tilde{g}_i) && \text{(variance of a sum)}\\
&= \underbrace{\beta_i^2\, \sigma_m^2}_{\text{market}} + \underbrace{\sigma^2_{\text{gen},i}}_{\text{idiosyncratic}}\ , && (\mathrm{Cov}(\tilde{g}_i, g_m) \approx 0)
\end{aligned} \tag{6}$$

Thus, the variance of g_i exceeds the per-asset (idiosyncratic) target $\mathrm{Var}(g_i) = \sigma^2_{\text{gen},i}$ by exactly $\beta_i^2\sigma_m^2$, an inflation that grows quadratically in the calibrated beta.

To remove the inflation, we deflate the generator draw before composing: form $\varepsilon_i := s_i\, \tilde{g}_i$ for some scalar $s_i \in (0, 1]$ and require $\mathrm{Var}(g_i) = \sigma^2_{\text{gen},i}$. Under the same independence assumption $\mathrm{Cov}(\tilde{g}_i,\, g_m) \approx 0$, $\mathrm{Var}(\beta_i\, g_m + s_i\, \tilde{g}_i) = \beta_i^2\, \sigma_m^2 + s_i^2\, \sigma^2_{\text{gen},i}$; setting this equal to $\sigma^2_{\text{gen},i}$ and solving yields the variance scale:

$$\boxed{s_i^2 \;=\; 1 \;-\; \frac{\beta_i^2\, \sigma_m^2}{\sigma^2_{\text{gen},i}}} \tag{7}$$

Define the *market loading ratio*:

$$\rho_i \;\equiv\; \frac{\beta_i^2\, \sigma_m^2}{\sigma^2_{\text{gen},i}}, \tag{8}$$

the share of the generator variance consumed by the market component; then $s_i^2 = 1 - \rho_i$, valid whenever $\rho_i < 1$, with the naive composition recovered in the $s_i = 1$ limit. Equation (7) breaks down when $\rho_i \geq 1$: the market loading alone would account for more than the entire generator variance, leaving the idiosyncratic share non-positive, a case that arises for high-β_i assets paired with low-volatility generators or whenever the synthetic market g_m is more volatile than the market on which β_i was calibrated. To handle this case, we imposed a floor $f \in (0, 1)$ on the idiosyncratic share by requiring $s_i^2 \geq f$, and clipped β_i downward whenever the floor was violated. Writing the clip threshold as $\rho_i^{\max} = 1 - f$, the piecewise clip rule is given by:

$$\beta_i^{\text{eff}} \;=\; \begin{cases} \beta_i & \text{if } \rho_i \leq \rho_i^{\max} \\ \mathrm{sign}(\beta_i)\sqrt{(1-f)\,\dfrac{\sigma^2_{\text{gen},i}}{\sigma_m^2}} & \text{if } \rho_i > \rho_i^{\max} \end{cases} \qquad s_i^2 \;=\; \begin{cases} 1-\rho_i & \text{if } \rho_i \leq \rho_i^{\max} \\ f & \text{if } \rho_i > \rho_i^{\max} \end{cases} \tag{9}$$

Here $\mathrm{sign}(\beta_i) \in \{-1, +1\}$ is the sign of the calibrated β_i, so the clipped slope inherits the calibrated direction; sign preservation keeps defensive (negative-β) assets defensive even when clipped. The floor f is a free parameter of the construction (values in Table 1); in our experiments we use $f = 0.10$, reserving at least 10% of each asset's variance for idiosyncratic noise.

Which target variance is appropriate depends on what the asset represents. For a typical single stock, the heavy tails, regime structure, and jumps live in the marginal the HMM-WJ generator was built to reproduce, so pinning $\mathrm{Var}(g_i)$ to $\sigma^2_{\text{gen},i}$ keeps the composition aligned with the per-asset calibration. Index-tracking ETFs reverse the picture: what distinguishes a tracker is its OLS relationship to the market, not its marginal. Against itself, SPY has $\beta = 1$, $R^2 = 1$, and zero residual variance exactly; against SPY, QQQ has $R^2 \approx 0.84$ and SPYG has $R^2 \approx 0.86$. A variance-preserving composition applied to a tracker supplies a residual whose variance bears no relation to the value $R^2_{i,\text{real}}$ implies, so OLS on the composed path returns $\hat{\beta}$ and $\hat{R}^2$ that drift away from their calibrated values. We therefore branch on the real-data calibration $R^2_{i,\text{real}}$ using a threshold R^2_{preserve} that separates index-tracking assets from single stocks: in our universe, index ETFs had $R^2_{i,\text{real}}$ above 0.80 and the most market-like single stocks had $R^2_{i,\text{real}}$ below 0.65 (Fig. S3), so we set $R^2_{\text{preserve}} = 0.80$ (Table 1). For any asset with $R^2_{i,\text{real}} \geq R^2_{\text{preserve}}$, inverting the population R^2 identity for Eq. (4) for the residual variance that hits the calibrated R^2 gives (Online Appendix S8):

$$\boxed{\sigma^2_{\varepsilon,\text{target}} \;=\; \beta_i^2\,\sigma_m^2 \cdot \frac{1-R^2_{i,\text{real}}}{R^2_{i,\text{real}}}} \tag{10}$$

The residual that achieves this target variance is the generator path $\tilde{g}_i$ rescaled by the ratio of target to generator standard deviation:

$$\varepsilon_i \;=\; \left(\sqrt{\frac{\sigma^2_{\varepsilon,\text{target}}}{\sigma^2_{\text{gen},i}}}\right)\tilde{g}_i, \tag{11}$$

which can then be copula-reordered and composed via (4). The construction recovers both calibrated quantities by design: $\hat{\beta} \to \beta_i$ in the large-T limit because ε_i is uncorrelated with g_m after the rank reorder, and $R^2 \to R^2_{i,\text{real}}$ by construction. When $R^2_{i,\text{real}} = 1$, Eq. (10) sets $\sigma^2_{\varepsilon,\text{target}} = 0$, recovering the SPY-against-itself identity $g_i = \alpha_i + \beta_i g_m$ without special-casing; conversely, when the rescale factor in Eq. (11) exceeds one (which happens when the synthetic market g_m has lower variance than the calibration market) the generator's tails are stretched to fit a target larger than the original draw, and we detect and flag the case so the loss of marginal fidelity is recorded.

Algorithm 2 summarizes the procedure: the calibrated $R^2_{i,\text{real}}$ selects a branch for each asset, applying the R^2-preserving construction of Eq. (10) when $R^2_{i,\text{real}} \geq R^2_{\text{preserve}}$ and the variance-preserving construction of Eq. (7) otherwise, clipped per Eq. (9) when the market loading ratio ρ_i would otherwise exceed $1-f$; the output records the branch used for each asset. The optional rank-reorder step accepts any joint dependence structure (empirical copula, t-copula, factor copula) because reordering preserves the marginal variance of ε_i, leaving the variance algebra of Eq. (7) unaffected. By construction the composed g_i satisfies three preservation criteria in the large-T limit: *SIM recovery*, where OLS of g_i on g_m returns $\hat{\alpha}_i = \alpha_i$ and $\hat{\beta}_i = \beta_i^{\text{eff}}$ (equal to β_i on the unclipped branch, attenuated by Eq. (9) otherwise); *marginal fidelity*, where $\text{Var}(g_i)$ matches the branch-selected target ($\sigma^2_{\text{gen},i}$ on the variance-preserving branch, or the value implied by the calibrated $R^2_{i,\text{real}}$ on the R^2-preserving branch); and *cross-sectional dependence*, where any joint dependence structure applied to $\{\varepsilon_i\}_{i=1}^N$ passes through unchanged to $\{g_i\}_{i=1}^N$ by the linearity of Eq. (4) in ε_i. The verification algebra, including the tail behavior implied by the floor f, is laid out in Online Appendix S8. The construction injects *contemporaneous* market dependence only; lagged dependence (lead-lag effects, market-driven volatility clustering) requires a richer factor structure than this construction provides.

We evaluated the composition on a 424-asset US equity and exchange-traded-fund universe drawn from Polygon.io daily-close histories, filtered to tickers with a complete in-sample trading history matching the reference length of AAPL ($T = 2{,}767$ daily closes covering January 2014 through December 2024); filtering on the common length preserved 424 tickers, of which the SPY total-return growth-rate series was taken as the market path g_m and the remaining 423 tickers as non-market assets. For each non-market ticker i we fitted a JumpHMM marginal to the asset's own growth-rate history using the same state-partition and emission configuration as the SPY analysis (Table 1; risk-free rate $r_f = 0$), holding the jump probability at $\epsilon = 0$ so each per-ticker generator reduced to the no-jump variant; this isolates the composition test from the temporal-clustering channel, which per-ticker jump tuning would otherwise add at the cost of a calibration step per asset. We then ran OLS for each ticker against the market path to obtain the calibrated SIM intercept, slope, coefficient of determination, and residual standard deviation; the universe spanned $\beta \in [0.01, 1.79]$ with median $\beta = 1.00$ and quartiles $(0.81, 1.00, 1.18)$, with $R^2_{i,\text{real}}$ ranging from 0.001 to 0.933 (Fig. S3).

We compared six composition methods on this universe. Naive plugged the JumpHMM draw directly into the SIM identity with no correction; Gaussian SIM substituted i.i.d. Gaussian noise scaled to the calibrated residual variance, recovering R^2 exactly but discarding marginal structure; Hybrid applied Algorithm 2 at the values in Table 1. A single innovation path drawn from each ticker's JumpHMM marginal was passed unchanged to Naive and Hybrid, a paired-innovation protocol that isolates the composition rule from generator stochasticity; Gaussian SIM, by construction, drew its own i.i.d. Gaussian innovations at the calibrated residual variance. Three further methods drew innovations

Algorithm 2 Hybrid SIM composition with variance correction.

Require: Market growth-rate path $g_m \in \mathbb{R}^T$ with sample variance σ_m^2; per-asset generator growth-rate paths $\tilde{g}_i \in \mathbb{R}^T$ for $i = 1, \dots, N$; calibrated SIM parameters $(\alpha_i, \beta_i, R^2_{i,\mathrm{real}})$; floor $f \in (0, 1)$; threshold $R^2_{\mathrm{preserve}} \in (0, 1)$.

Ensure: Composed paths $g_i \in \mathbb{R}^T$ and per-asset construction flags.

for each asset $i = 1, \dots, N$ **do**
  Compute generator variance $\sigma^2_{\mathrm{gen},i} \leftarrow \mathrm{Var}(\tilde{g}_i)$.
  if $R^2_{i,\mathrm{real}} \geq R^2_{\mathrm{preserve}}$ **then**
    $\sigma^2_{\varepsilon,\mathrm{target}} \leftarrow \beta_i^2 \sigma_m^2 (1 - R^2_{i,\mathrm{real}}) / R^2_{i,\mathrm{real}}$ {Eq. (10)}
    $\varepsilon_i \leftarrow \left(\sqrt{\sigma^2_{\varepsilon,\mathrm{target}} / \sigma^2_{\mathrm{gen},i}} \right) \tilde{g}_i$
    $\beta_i^{\mathrm{eff}} \leftarrow \beta_i$; branch $\leftarrow$ R^2-preserving
  else
    $\rho_i \leftarrow \beta_i^2 \sigma_m^2 / \sigma^2_{\mathrm{gen},i}$ {Eq. (8)}
    if $\rho_i \leq 1 - f$ **then**
      $s_i^2 \leftarrow 1 - \rho_i$; $\beta_i^{\mathrm{eff}} \leftarrow \beta_i$; branch $\leftarrow$ variance-preserving
    else
      $s_i^2 \leftarrow f$; $\beta_i^{\mathrm{eff}} \leftarrow \mathrm{sign}(\beta_i) \sqrt{(1 - f)\, \sigma^2_{\mathrm{gen},i} / \sigma_m^2}$; branch $\leftarrow$ clipped variance-preserving
    end if
    $\varepsilon_i \leftarrow s_i \, \tilde{g}_i$
  end if
  (Optional) apply cross-sectional rank reorder to ε_i.
  $g_i \leftarrow \alpha_i + \beta_i^{\mathrm{eff}} g_m + \varepsilon_i$ {Eq. (4)}
end for

Table 1: **HMM-WJ and SIM composition hyperparameters.** Values used for the main SPY analysis, with the sensitivity-sweep or grid-search span and the location of the corresponding check where one exists. The 423-ticker SIM composition universe holds $\epsilon = 0$ instead of the SPY value; per-ticker (ϵ^*, λ^*) for the NVDA, JNJ, and JPM cross-asset validation are given in Table S7.

Symbol	**Description**	**SPY value**	**Explored range**
State partition and emission			
N	Laplace-quantile states	100	$\{30, \dots, 350\}$, Online Appendix S6
ν	Student-t emission degrees of freedom	5	$\{3, \dots, 30, \infty\}$
Jump-duration override			
ϵ	jump probability per step	10^{-4}	$[10^{-4}, 2.5 \times 10^{-2}]$, Eq. (3)
λ	mean jump-episode duration (steps)	100	$[10, 160]$, Eq. (3)
p_{neg}	probability a jump targets $\mathcal{S}_-$	0.52	fixed
N_{tail}	quantile states per tail set	5	fixed (at $N = 100$)
w_κ	kurtosis-penalty weight, Eq. (3)	0.20	fixed
L	ACF lag window, Eq. (3)	25 days	fixed
SIM composition			
f	idiosyncratic variance floor	0.10	$\{0.05, \dots, 0.30\}$, Fig. S7
R^2_{preserve}	branch-selection threshold	0.80	$\{0.70, \dots, 0.90\}$, Fig. S7

from residual generators fit directly to the OLS residual series: JumpHMM-on-residuals converted each ticker's residual series into a synthetic price path by cumulative compounding and fitted a fresh JumpHMM marginal to that path; block bootstrap [52] used Politis–Romano stationary bootstrap replicates with mean block length $\sqrt{T} \approx 50$; and GARCH(1,1)-t [7] fitted a conditional-variance model per ticker (30 of 423 non-stationary fits excluded from that method only). We skipped the optional cross-sectional rank reorder of Algorithm 2 for all six methods so per-ticker metrics reflected the marginal construction alone.

Table 2: **In-sample and out-of-sample model comparison for SPY** (1,000 simulated paths, significance level $\alpha = 0.05$). Bootstrap: i.i.d. resample of training data. Gaussian/Laplace: i.i.d. draws from distributions fitted by maximum likelihood estimation (MLE). GARCH: GARCH(1,1) estimated by quasi-maximum likelihood. GRU: 2-layer gated recurrent unit [53] with Gaussian output head. HSMM: hidden semi-Markov model with $K = 8$ Laplace quantile states, negative-binomial dwell times, and Student-t ($\nu = 5$) emissions [16]. HMM-NJ: hidden Markov model ($N = 100$ states) without jumps. HMM-WJ: hybrid model with Poisson jump-duration extension. ACF-MAE: mean absolute error of the ACF of $|G_t|$ over lags 1–252. Wasserstein-1 and Hellinger distance defined in Online Appendix S3. Coverage: fraction of 99 empirical quantiles (1st–99th) falling within the [5th, 95th] percentile envelope of the corresponding synthetic quantiles. Standard errors (SE) in parentheses: binomial SE for pass rates, $\mathrm{std}/\sqrt{n}$ for kurtosis, Wasserstein-1, and Hellinger; bootstrap SE ($B = 500$) for ACF-MAE and coverage. Best value per row in bold.

Metric	**Bootstrap**	**Gaussian**	**Laplace**	**GARCH(1,1)**	**GRU**	**HSMM**	**HMM-NJ**	**HMM-WJ**
In-sample: 2,766 *trading days* (2014–2024)								
KS pass rate (%) ↑	100.0 (<0.1)	0.0 (<0.1)	44.0 (1.6)	5.5 (0.7)	0.6 (0.2)	82.0 (1.2)	**99.7** (0.2)	97.6 (0.5)
AD pass rate (%) ↑	99.7 (0.2)	0.0 (<0.1)	43.3 (1.6)	1.9 (0.4)	0.2 (0.1)	42.5 (1.6)	**99.1** (0.3)	91.3 (0.9)
Excess kurtosis (observed)				7.715				
Excess kurtosis (simulated) ≈	7.6 (0.08)	−0.0 (<0.01)	3.0 (0.02)	**8.2** (0.37)	5.7 (0.16)	4.8 (0.08)	8.1 (0.14)	7.6 (0.14)
ACF-MAE ↓	0.060 (<0.001)	0.060 (<0.001)	0.060 (<0.001)	**0.031** (0.001)	0.036 (<0.001)	0.059 (<0.001)	0.059 (<0.001)	0.052 (<0.001)
Coverage (%) ↑	100.0 (<0.1)	13.1 (0.1)	37.4 (1.5)	29.3 (1.4)	17.2 (0.5)	68.7 (1.1)	**100.0** (<0.1)	**100.0** (<0.1)
Wasserstein-1 ↓	0.062 (0.001)	0.399 (0.001)	0.138 (0.001)	0.304 (0.006)	0.421 (0.003)	0.176 (0.001)	**0.081** (0.001)	0.101 (0.001)
Hellinger dist ↓	0.042 (<0.001)	0.148 (<0.001)	**0.072** (<0.001)	0.100 (0.001)	0.134 (0.001)	0.113 (<0.001)	0.073 (<0.001)	0.075 (<0.001)
Out-of-sample: 249 *trading days* (2025)								
KS pass rate (%) ↑	97.4 (0.5)	62.2 (1.5)	88.0 (1.0)	80.3 (1.3)	71.8 (1.4)	96.2 (0.6)	**96.7** (0.6)	94.4 (0.7)
AD pass rate (%) ↑	98.5 (0.4)	46.3 (1.6)	92.9 (0.8)	72.0 (1.4)	44.8 (1.6)	**96.7** (0.6)	**96.7** (0.6)	95.1 (0.7)
Excess kurtosis (observed)				6.867				
Excess kurtosis (simulated) ≈	6.0 (0.18)	−0.0 (<0.01)	2.7 (0.05)	1.6 (0.07)	2.9 (0.10)	4.1 (0.13)	**6.4** (0.22)	6.1 (0.13)
ACF-MAE ↓	0.043 (<0.001)	0.043 (<0.001)	0.043 (<0.001)	**0.026** (<0.001)	0.033 (<0.001)	0.042 (<0.001)	0.041 (<0.001)	0.039 (<0.001)
Coverage (%) ↑	100.0 (<0.1)	36.4 (1.9)	74.7 (1.0)	96.0 (1.3)	77.8 (2.9)	**100.0** (0.2)	**100.0** (<0.1)	**100.0** (<0.1)
Wasserstein-1 ↓	0.232 (0.002)	0.452 (0.002)	0.263 (0.002)	0.507 (0.018)	0.488 (0.005)	0.287 (0.002)	**0.258** (0.002)	0.282 (0.006)
Hellinger dist ↓	0.205 (0.001)	0.235 (0.001)	0.211 (0.001)	0.232 (0.001)	0.240 (0.001)	0.239 (0.001)	**0.207** (0.001)	0.210 (0.001)

For every composed series we computed three metric families against the ticker's real growth-rate history (definitions in Online Appendix S3): SIM recovery via OLS on the market path, reporting the absolute deviations $|\hat{\beta} - \beta|$ and $|\hat{R}^2 - R^2_{\text{real}}|$; marginal fidelity via Kolmogorov–Smirnov (KS) and Anderson–Darling (AD) two-sample p-values against the real marginal, plus Wasserstein-1 distance, excess kurtosis, and the Hill upper-tail index on the upper 5% order statistics of $|g|$; and variance preservation via the ratio of composed to generator sample variance. We report pass rates at $\alpha = 0.05$ with $R = 100$ replications per ticker (42,300 per-ticker observations per method) and a fixed simulation seed of 1234.

4 Results

4.1 Single-asset fit and temporal tradeoff

We first measured the empirical SPY data targets that any synthetic generator must reproduce. Descriptive statistics on the real SPY series confirmed all three stylized facts in both the in-sample (2014–2024, $T = 2{,}766$) and out-of-sample (2025, $T = 249$) windows (Table S1): excess kurtosis with Jarque-Bera rejection, persistent volatility clustering, and near-zero linear autocorrelation in raw returns [29].

To meet those empirical targets, we calibrated (ϵ, λ) for the HMM-WJ simulator (Algorithm 1) via the multi-objective grid search of Eq. (3) against the observed in-sample SPY series (search range in Table 1). Several grid points had similar objective values, so we interpret $(\epsilon, \lambda) = (10^{-4}, 100)$ as one suitable setting rather than a unique optimum, one that delivered strong distributional and temporal fidelity on SPY (Table 2): at these values, the ACF of $|G_t|$ closely tracked the observed SPY ACF across lags 1–252, indicating that the jump-duration override partially reproduced the empirical ARCH effect. The primary analysis used $N = 100$ states, but a sensitivity sweep (range in Table 1) confirmed robustness up to $N = 200$: KS and AD pass rates remained stable for both HMM-NJ (without jumps) and HMM-WJ (with jumps), with every state receiving adequate statistical support. At $N = 350$, certain states were never visited in the historical record, setting a practical upper bound on state resolution (Online Appendix S6).

With the model calibrated, we benchmarked HMM-WJ (Algorithm 1) against seven alternative generators on 1,000 synthetic paths of 2,766 trading days each (Table 2, Figure 3). Bootstrap

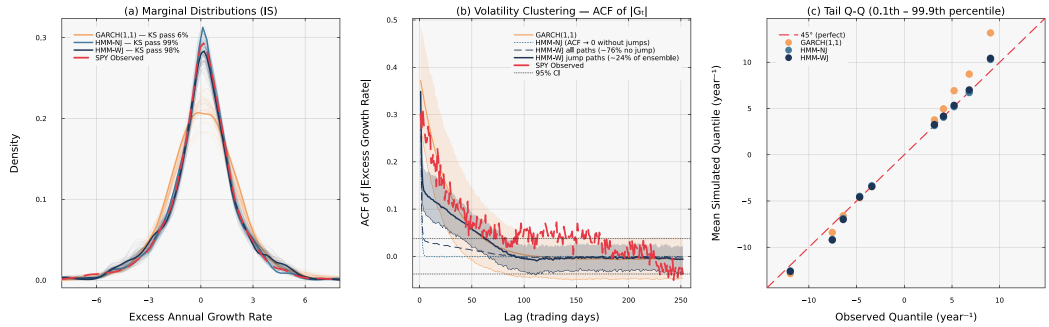

Figure 3: **Head-to-head in-sample model comparison for SPY** ($N = 100$, 1,000 simulated paths). *Panel (a):* marginal density of excess growth rates with in-sample KS pass rates annotated per generator. *Panel (b):* autocorrelation function of $|G_t|$ at lags 1–252 (shaded bands: 10th–90th percentile across paths) per generator. *Panel (c):* tail Q-Q plot at the 0.1st–99.9th percentile region per generator.

resampling had the best marginal fit (100% KS, 100% coverage); Gaussian sampling had the worst (0% KS). Every other parametric and neural baseline excelled on one quality dimension but failed on another. GARCH(1,1) achieved the best temporal fidelity (ACF-MAE 0.031) but a poor distributional fit (5.5% KS). The Gated Recurrent Unit (GRU) [53] achieved an ACF-MAE of 0.036 but a 0.6% KS pass rate. The hidden semi-Markov model (HSMM) [16] reached 82% KS but its coarse 8-state partition left simulated kurtosis at 4.8 against an observed 7.7 and only 68.7% quantile coverage. A head-to-head comparison isolating the persistence design (HMM-WJ vs HSMM) and emission family (Gaussian vs Student-t) is given in Table S3 (Sec. S5). Both HMM variants delivered the highest distributional fidelity among parametric models. HMM-WJ's pass rates ran 2–8 percentage points below HMM-NJ's because jumps occasionally overweight tail states relative to their empirical frequency. The Student-t ($\nu = 5$) emission held HMM-WJ's simulated kurtosis close to the observed value (7.6 against 7.7), while Normal emissions underestimated it by $\sim 29\%$ (Table S3). Negative skewness (-0.75 observed, Table S1) was reproduced by both HMM variants, arising from the asymmetric state-occupancy of the Laplace partition rather than any explicit skewness parameter; the symmetric-innovation models (GARCH, Gaussian, Laplace) produced near-zero skewness by construction. On the temporal dimension, HMM-WJ reduced ACF-MAE from 0.059 for HMM-NJ to 0.052; GARCH still had the lowest ACF-MAE at 0.031, and only HMM-WJ, among the non-GARCH, non-neural models, scored below the shared i.i.d. value of 0.060, driven by the $\sim 24\%$ of paths that contained at least one Poisson jump and sustained $\mathrm{ACF}(|G_t|)$ decay across all lags (the remaining 76% behaved identically to HMM-NJ). Out-of-sample on the held-out 249 days of 2025 the same ranking held (Table 2, Figure S4): HMM-NJ and HMM-WJ achieved 96.7% and 94.4% KS with simulated kurtosis of 6.4 and 6.1 against an observed 6.9, while GARCH's W_1 deteriorated by 67% from in-sample, exposing a tail misspecification that binary pass rates obscured. In the extreme tails, HMM-WJ's simulated quantiles matched the empirical tail more closely than the other seven generators (Fig. 3c). Cross-asset evaluation on NVDA, JNJ, and JPM produced stable in-sample fits with HMM-WJ KS $\geq 98.9\%$ on all three; the out-of-sample pass rate held at 97.7% for NVDA and attenuated to 74.0 and 79.2% for JNJ and JPM, reflecting larger between-window distributional shifts on the defensive and financial tickers than on either the SPY ETF or the semiconductor name (Table S7, Online Appendix S7).

Having traced HMM-WJ's temporal gain to jump-active paths, we resampled the fitted ensemble to a target jump-path fraction f and measured the in-sample stylized facts as a function of the mix, to characterize the temporal-marginal tradeoff (Figure 4). The jump-path fraction f is the share of accepted paths containing at least one jump; because that share is governed by the jump probability ϵ rather than fixed directly, we swept f from 0 (accepted paths reduce to the no-jump generator HMM-NJ) to 1 (every accepted path contains a jump), holding the fitted model fixed and reweighting only the accepted mixture. The absolute-return autocorrelation error fell monotonically from 0.059 at $f = 0$ to 0.037 at $f = 1$, moving from the shared i.i.d. value toward the GARCH value as the jump share rose. Over the same sweep the simulated excess kurtosis declined from 7.9 to 6.7, matching the observed value near $f \approx 0.14$ and falling below it thereafter, and the two-sample KS and AD marginal pass rates declined in step. The Hill upper-tail index stayed flat (ξ near 0.35 against an observed

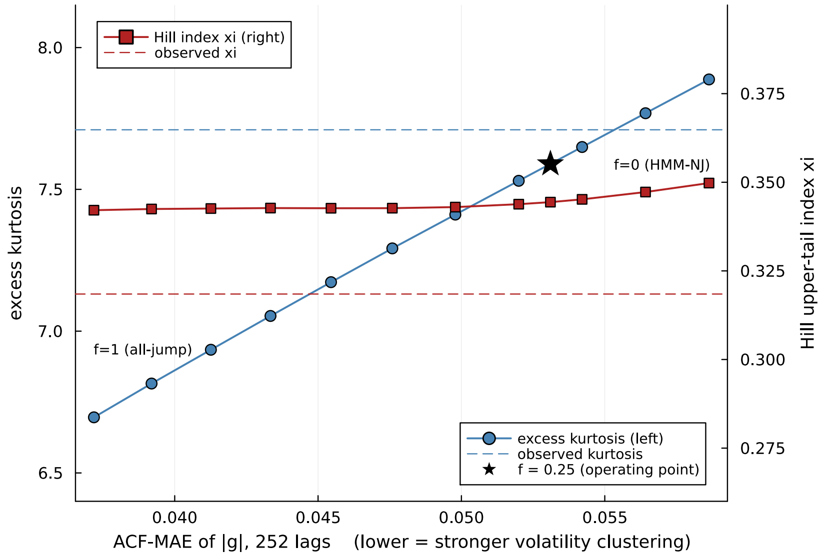

Figure 4: **Jump-mix enrichment frontier for SPY** (in-sample, $N = 100$, pool of $4{,}000$ simulated paths). Resampling the fitted HMM-WJ ensemble to a target jump-path fraction f (the share of accepted paths containing at least one jump) traces the temporal-marginal tradeoff. Horizontal axis: absolute-return autocorrelation error (ACF-MAE of $|G_t|$ over lags 1–252). Left axis (blue): simulated excess kurtosis. Right axis (red): Hill upper-tail index ξ, the raw estimator equal to the reciprocal of the tail exponent. Dashed lines mark the observed SPY values; the star marks the $f \approx 0.25$ value used throughout. Endpoints $f = 0$ and $f = 1$ correspond to the no-jump generator and an all-jump ensemble.

Table 3: **Aggregate scorecard across** 423 **non-market tickers and** 100 **replications per ticker.** Median values per composition method. $|\hat{\beta} - \beta|$ and $|\hat{R}^2 - R^2_{\text{real}}|$: absolute deviation between OLS recovery and calibration. KS pass and AD pass: fraction of replications whose two-sample test against the real marginal exceeds $p = 0.05$. W_1: Wasserstein-1 distance to the real marginal. κ: sample excess kurtosis. Hill: tail-index estimate on the upper 5% of $|g|$.

Composer	$\lvert\hat{\beta} - \beta\rvert$	$\lvert\hat{R}^2 - R^2_{\text{real}}\rvert$	KS pass (%)	AD pass (%)	W_1	κ	Hill
Naive	0.022	0.087	4.5	2.7	0.710	7.767	0.369
Gaussian SIM	0.017	0.008	0.6	0.5	0.682	1.427	0.217
Hybrid	0.018	0.021	50.4	47.6	0.272	7.165	0.365
JumpHMM-on-residuals	0.018	0.015	75.8	70.7	0.232	7.262	0.365
Block bootstrap	0.017	0.020	73.3	66.8	0.232	6.675	0.330
GARCH(1,1)-t	0.017	0.036	67.4	61.4	0.267	6.069	0.317

0.32) across the whole sweep: the Student-t ($\nu = 5$) emission sets the tail exponent independently of jump content, so enriching the jump population trades volatility clustering against distributional body fidelity rather than tail weight. The value $f \approx 0.25$ used throughout was chosen where the simulated kurtosis remained close to the observed value while the autocorrelation error had already dropped below the i.i.d. level.

4.2 Multi-asset composition

With the per-asset marginals validated on single-asset stylized facts in both windows, we tested whether SIM composition preserved each asset's variance and factor loading across the 423-ticker universe, comparing the six composition methods of Table 3. Every method recovered β to a similarly tight tolerance, as expected since each respects the SIM identity in expectation: the median absolute error ranged from 0.018 to 0.022 across methods. The methods differed in R^2 recovery, marginal fit, and tail behavior. Gaussian SIM recovered R^2 most tightly (median error 0.008) because it targets R^2 by construction, but it failed marginal fidelity badly: KS pass rate 0.6%, AD pass rate 0.5%, excess kurtosis 1.4 (real data, ~ 13), Hill tail index 0.22 (versus ~ 0.37 on the generator; Fig. S2). Naive composition retained the generator marginal but inflated its variance by $\beta_i^2\sigma_m^2$, so its KS and AD pass rates were 4.5% and 2.7%. Hybrid held both properties at once: KS pass rate 50.4%, AD pass rate 47.6%, kurtosis 7.2, Hill 0.36; its R^2 recovery (0.021) came in tighter than naive but looser than Gaussian SIM, reflecting that the variance-preserving branch targets $\sigma^2_{\mathrm{gen},i}$ rather than $R^2_{i,\mathrm{real}}$. The three residual-fit methods sidestepped the variance-inflation problem entirely by fitting a generator directly to the empirical OLS residual series e_i, whose sample variance $\sigma^2_{\varepsilon,\mathrm{real},i}$ excludes the market component by construction; their aggregate KS pass rates were 75.8% (JumpHMM-on-residuals), 73.3% (block bootstrap), and 66.9% (GARCH(1,1)-t), each higher than hybrid's 50.4%. Residual-fit methods improved the center of the distribution but had similar tail estimates: hybrid's median excess kurtosis (7.17) and Hill upper-tail index (0.365) matched JumpHMM-on-residuals (7.26 and 0.365) to within sampling variability, confirming that the variance correction of Eq. (7) preserves the heavy-tailed character of the full-return generator even when the rescaling is applied; block bootstrap and GARCH, by fitting their own residual generators, reproduced the empirical residual series but with slightly lighter tails ($\kappa = 6.68$ and 6.04; Hill 0.330 and 0.318) than either JumpHMM-based method.

4.3 VaR coverage

To test whether the distributional differences affected Value-at-Risk (VaR) coverage, we ran a per-ticker one-day VaR coverage check on the same universe (Table 4). At $\alpha = 0.99$ the nominal exceedance rate is 1%; naive composition's empirical exceedance rate fell short of nominal at 0.67%, while Gaussian SIM's exceeded it at 1.40%, both failing Kupiec unconditional coverage on the majority of the universe (45 and 47% pass rates), while hybrid, JumpHMM-on-residuals, block bootstrap, and GARCH(1,1)-t all hit coverage within one-tenth of a percentage point of nominal (0.95, 0.99, 1.09, and 1.09% respectively), with Kupiec pass rates clustered between 72 and 86%. Hybrid and JumpHMM-on-residuals differed by less than the cross-ticker spread on this test: the 25 percentage-point difference in KS pass rate between them (Table 3) did not translate into a material VaR coverage difference. Hybrid matched a dedicated residual generator on tail accuracy when the per-asset generator was specified in full-return units, without requiring a separate fitting stage.

4.4 Sensitivity analyses

Having established that body-fidelity did not translate into VaR accuracy, we compared per-ticker KS pass rate with the per-ticker marginal variance ratio $\mathrm{Var}(g)/\sigma^2_{\mathrm{gen}}$ against the calibrated β (Fig. 5). The variance-ratio panel showed the cause directly: the naive scatter followed the theoretical curve $1 + \rho_i$, overshooting generator variance by 50–75% at high β, while hybrid held the variance ratio

Table 4: **Per-ticker one-day Value-at-Risk coverage check across the six composition methods.** For each (ticker, replication) pair we estimate the VaR threshold at coverage level α from the composed path and count exceedances against the real 2014–2024 growth-rate history. *rate*: mean exceedance rate across tickers (nominal: $1 - \alpha$). *SD*: cross-ticker standard deviation of the exceedance rate, in percentage points. *Kupiec pass*: fraction of tickers on which the Kupiec unconditional-coverage test p-value exceeds 0.05. Hybrid and JumpHMM-on-residuals differ by less than the cross-ticker spread at both coverage levels; naive composition and Gaussian SIM fail coverage on a majority of the universe.

	$\alpha = 0.95$			$\alpha = 0.99$		
Composer	rate (%)	SD (pp)	Kupiec pass (%)	rate (%)	SD (pp)	Kupiec pass (%)
Naive	3.55	0.52	10.4	0.67	0.23	45.2
Gaussian SIM	4.06	0.76	42.6	1.40	0.29	47.0
Hybrid	4.75	0.56	80.4	0.95	0.27	79.7
JumpHMM-on-residuals	4.87	0.55	84.0	0.99	0.26	82.9
Block bootstrap	4.78	0.64	76.7	1.09	0.25	86.0
GARCH(1,1)-t	4.64	0.93	69.0	1.09	0.35	72.1

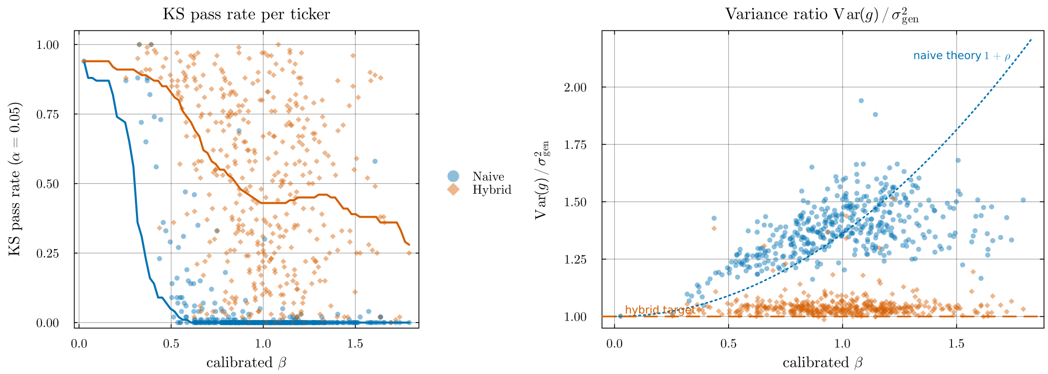

Figure 5: **Hybrid's variance correction and its KS-pass-rate consequence, per ticker vs calibrated β.** Each point is a per-ticker median over 100 replications; thick lines are binned medians. *Left (consequence):* Kolmogorov–Smirnov pass rate against the real marginal at $\alpha = 0.05$ for naive composition (blue) and hybrid (orange). *Right (cause):* variance ratio $\mathrm{Var}(g)/\sigma^2_{\text{gen}}$ for the same two methods. The dotted curve is the theoretical naive inflation $1 + \rho$ evaluated at the universe-median σ^2_{gen}; the dashed line marks a variance ratio of one.

near one by construction. The KS-pass-rate panel showed the consequence: naive fell from ~ 0.25 at $\beta \approx 0.3$ to below 0.01 past $\beta \approx 0.7$, while hybrid remained between 0.3 and 0.9 across the full range. Applying the branch-selection rule of Algorithm 2 to the 423-ticker universe assigned nearly every ticker to the variance-preserving branch, with only two trackers routed to the R^2-preserving branch and none clipped (Fig. S6, Table S4). QQQ ($R^2 = 0.861$) and SPYG ($R^2 = 0.933$) were the two R^2-preserving assignments, both passing KS at 99% since the branch recovers both β and R^2 by construction; the highest-R^2 individual stock (BLK at $R^2 = 0.645$) fell ~ 0.16 below the $R^2_{\text{preserve}} = 0.80$ threshold, so the assignment was empirically tracker-specific on this universe rather than a threshold artifact. Clipping did not trigger anywhere in the universe: $\rho_i = \beta_i^2 \sigma_m^2 / \sigma^2_{\text{gen},i}$ (Eq. (8)) stayed below $1 - f = 0.90$ for every ticker because the JumpHMM marginals had enough variance to keep the market component below the $f = 0.10$ idiosyncratic floor. Because the R^2-preserving branch had only two empirical assignments, we validated it on a synthetic-tracker grid spanning $(\beta_{\text{true}}, R^2_{\text{true}}) \in \{0.8, 1.0, 1.2\} \times \{0.80, 0.85, 0.90, 0.95, 0.99\}$; the branch selector recovered the target $\hat{\beta}$ and $\hat{R}^2$ within sampling tolerance and passed KS on every cell.

Bucketing the universe by β quartile, the hybrid KS pass rate declined gently across the range (Table S5). The rate fell from 67% at low β to 44% at high β as ρ_i approaches the clipping threshold, tightening the market-driven component of g toward Gaussian along its own axis; hybrid still beat both baselines by more than an order of magnitude in every bucket. The same clipping mechanism was stress-tested directly by rescaling the market growth-rate path by $\gamma \in \{1, 2, 3\}$, holding the calibrated marginals fixed (Table S6). At $\gamma = 1$ no ticker crossed the threshold and hybrid KS matched the unstressed value (50.3%); at $\gamma = 2, 3$ the clipping branch activated on 76.4% and 95.2% of tickers and hybrid KS rose to 71.8% and 77.3%: under stress, clipping increased the KS pass rate rather than reducing it.

Finally, we tested the aggregate hybrid KS pass rate of Table 3 against the two hyperparameter choices of Algorithm 2 (Table 1), re-running hybrid on the full 423-ticker universe at every point of the $R^2_{\text{preserve}} \times f$ grid in Fig. S7. The hybrid KS pass rate ranged from 50.24% to 50.38% across the 25 grid cells, a spread of 0.14 percentage points; the R^2-preserve branch assignment stayed fixed at QQQ and SPYG for every threshold at or below $R^2 = 0.85$, and the floor f did not change any branch assignment across the grid. The aggregate pass rate was therefore invariant to the hyperparameter values within wide tolerances. We then quantified the contribution of generator stochasticity to the aggregate metrics by re-running the full six-method protocol under 8 random seeds (Table S8). Per-method KS pass rates held to within fractions of a percentage point across seeds, and the seed sweep reproduced the relative ordering of the methods without inversions. The across-method spread therefore exceeded the seed-induced spread by roughly two orders of magnitude, confirming that the differences between hybrid, the baselines, and the residual-fit methods are not artifacts of the simulation seed.

5 Discussion

In this study, we found that HMM-WJ improved volatility persistence relative to HMM-NJ but slightly reduced marginal fit, with pass rates 2–8 percentage points lower in-sample. Among the alternatives, each excelled on one property and failed on another for a reason tied to its own structure: GARCH's conditional-variance dynamics reproduced temporal clustering but not the marginal's body; the neural baseline matched some of the temporal structure despite carrying tens of thousands of parameters, yet still missed the marginal; the semi-Markov baseline's coarse partition left it unable to span the full growth-rate range; and bootstrap resampling passed every distributional test by construction, since it resamples only observed returns rather than extrapolating beyond the historical record. Among these alternatives, HMM-WJ was the only non-GARCH, non-neural model to reduce the autocorrelation of absolute returns below the shared i.i.d. level while retaining most of HMM-NJ's distributional fit, an ordering that held out-of-sample on the 2025 evaluation. The model therefore offers a tradeoff rather than a uniformly better result.

The pass-rate comparisons come with a statistical caveat: two-sample KS and AD p-values assume independent observations, an assumption the HMM, HSMM, GARCH, and bootstrap generators all violate, so we weigh pass rates alongside the distribution-level distances (W_1, Hellinger, and the Hill tail index) rather than treat them as calibrated on a common footing across generators. Three further caveats apply to the per-asset construction. The $N = 100$ state resolution, selected by a sensitivity sweep over $\{30, 60, 90, 100, 150, 200\}$, worked well for the 2,766-day SPY sample, but shorter histories will support fewer states before empirical visits per state thin out. The jump parameters (ϵ, λ) were tuned once against SPY and applied across the universe; per-ticker tuning on the three cross-asset tickers produced strong in-sample fits (KS $\geq 98.9\%$, Online Appendix S7), but is not guaranteed to transfer to assets with different jump behavior without its own calibration step. The Student-t emission with $\nu = 5$ improved kurtosis fidelity over Normal emissions but does not model skewness directly: the negative skewness both HMM variants reproduce comes from the asymmetric state-occupancy of the Laplace partition, so assets whose skew direction the partition cannot induce will not be matched without adjusting the quantile rule.

We then evaluated the multi-asset extension. The variance correction reduced the composed marginal's variance error and improved KS pass rates relative to naive composition, which inflates variance by $\beta_i^2\sigma_m^2$, and relative to Gaussian SIM, which recovers β exactly but strips the marginal entirely. The three residual-fit methods still had better marginal fit than hybrid (67 to 76% KS pass rate versus 50%), because fitting a generator directly to the OLS residual series avoids the variance-inflation problem rather than correcting for it after the fact; the choice between the two approaches is governed by whether the regime-switching and jump structure of the full-return marginal is itself a property the per-asset generator should have, for instance when the same generator is used for both single-asset and multi-asset purposes. On tail risk this tradeoff disappears: hybrid and the three residual-fit methods all matched on 99% VaR coverage, differing by less than the cross-ticker spread, so hybrid delivers tail accuracy on par with a dedicated residual generator without a separate fitting stage, even though its marginal body fit trails the residual-fit alternatives.

Several evaluation choices bound these results. We fixed the market path at the real SPY history, paired Naive and Hybrid on the same per-ticker JumpHMM innovation path so the comparison isolates the composition rule (Gaussian SIM and the three residual-fit methods each drew from their own generators), and skipped the optional cross-sectional rank reorder; we also evaluated VaR per ticker rather than at the portfolio level, which the optional reorder would be needed to support. The variance correction assumes $\mathrm{Cov}(\tilde{g}_i, g_m) \approx 0$, which holds by construction here since the JumpHMM marginals are fit and sampled independently of the market path, but would need adjustment for a generator already correlated with the market (for example, a bootstrap from joint history). The construction also injects contemporaneous market dependence only: lagged effects such as lead-lag correlation or market-driven volatility clustering are not reproduced. Clipping did not trigger anywhere in this universe because the JumpHMM marginals had enough variance to keep every ticker's market loading below the idiosyncratic floor; it becomes relevant for scenario-driven markets more volatile than the calibration history, or for universes mixing low-volatility stocks with extreme-beta assets such as leveraged ETFs. A related, milder attenuation appeared even without clipping: the hybrid KS pass rate fell from 67% at low β to 44% at high β, because as the market-driven component approaches the idiosyncratic floor it accounts for an increasing share of the marginal, and that component is only as heavy-tailed as the market path itself.

Independent residuals produce a diversification-return bias under daily rebalancing: matching each asset's marginal variance does not restore the contemporaneous cross-asset residual covariance that common sector and style factors induce in real markets [46], so independent residuals overstate the daily cross-sectional dispersion that rebalancing compounds into a spurious return premium. This affects every i.i.d.-residual composition method evaluated here, not only naive composition; block bootstrap and GARCH(1,1)-t restore each ticker's own temporal structure but not the cross-asset covariance, so the bias remains even for the methods that best match each ticker's marginal. In a 20-asset target-weight allocator example, the synthetic median annualized growth rate exceeded the real 2025 return by about 22 percentage points, with a static-versus-daily-rebalance decomposition consistent with a diversification-return explanation rather than a calibration error [46, 54, 55] (Online Appendix S2). We applied a location-shift correction that subtracts a constant drift in log-wealth from every path, calibrated to match an external long-run prior [56, 57] rather than the realized test-year outcome. The correction preserves the rank order of paths and the volatility envelope, but because the drift is time-dependent in wealth space it does move drawdown dates and depths, so it recalibrates the ensemble level rather than restoring the missing cross-asset covariance. Full methodology, the rejected alternatives, and the percentile placement of the realized year are given in Online Appendix S2.

Four extensions remain open. Per-ticker tuning of the jump parameters (ϵ, λ) on each asset's own history would likely raise marginal pass rates at the cost of a calibration step per asset. Multi-factor and lagged-factor single-index variants would replace the single market factor with K factors or lags, capturing the lagged market dependence the current construction omits; the variance correction generalizes to both without changing its algebra, though the empirical evaluation is left to follow-up work. A jointly specified residual model, such as a synchronized multivariate block bootstrap, a residual factor model, or a copula-based dependence model, composed with the single-index factor instead of independent per-asset draws, would restore cross-asset residual covariance and address the diversification-return bias directly; an ordinary per-ticker autoregressive or block-bootstrap residual model would still be fit and sampled independently for each ticker, so it would improve each asset's own temporal dependence without creating dependence between assets. A stress-test harness that pairs the clipping rule with a market scenario more volatile than the calibration history would test whether the idiosyncratic floor keeps per-asset marginals heavy-tailed under exactly the conditions where naive SIM composition breaks down.

6 Conclusion

In this study, we developed a hybrid hidden Markov model for synthetic equity path generation that discretizes excess growth rates into Laplace-quantile states with Student-t emissions and a Poisson jump-duration override. We estimated the transition matrix by direct frequentist counting rather than the iterative Baum-Welch algorithm. For SPY, the jump-duration override reduced absolute-return autocorrelation error from 0.059 to 0.052 relative to the no-jump variant, the only non-GARCH, non-neural model to score below the shared i.i.d. value, while GARCH retained the lowest error at 0.031. HMM-WJ is therefore a tradeoff between marginal and temporal fit, passing the Kolmogorov-Smirnov test against the real SPY distribution at 97.6% in-sample and 94.4% out-of-sample.

We also derived a closed-form scale correction for single-index composition that removes the variance inflation caused by inserting a per-asset generator draw into the market factor. The correction recovered each asset's calibrated factor loading to a median absolute error of 0.018 across 423 non-market assets and matched the leading residual-fit methods on a per-ticker one-day 99% Value-at-Risk coverage check, while retaining heavier tails than the residual-fit methods. The current multi-asset implementation does not model cross-asset residual covariance: in a deployed 20-asset allocator, the synthetic median daily-rebalanced return ran about 22 percentage points above the realized 2025 return, consistent with a diversification-return effect from independent per-asset residuals. A location-shift correction set to a documented long-run prior offsets this gap without restoring the missing covariance, so a full joint residual model is needed before using these paths for portfolio-level return forecasts.

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Author contributions. A.A. designed the hybrid hidden Markov marginal generator, performed the single-asset experiments and validation framework, and contributed to the manuscript. J.D.V. designed the variance-corrected single-index composition, performed the multi-asset experiments and downstream Value-at-Risk validation, developed the bias-correction methodology, released the code, and contributed to the manuscript.

Competing interests. The authors declare no competing interests.

Code and data availability. The experiment scripts, evaluation harness (six-composer benchmark, Value-at-Risk coverage check, bias-correction utility), and the fitted per-asset marginals are released in the paper repository (`https://github.com/varnerlab/HMM-w-jumps-paper`); the underlying model-fitting library is released separately as `JumpHMM.jl` (`https://github.com/varnerlab/JumpHMM.jl`), pinned as a dependency of the experiment code.

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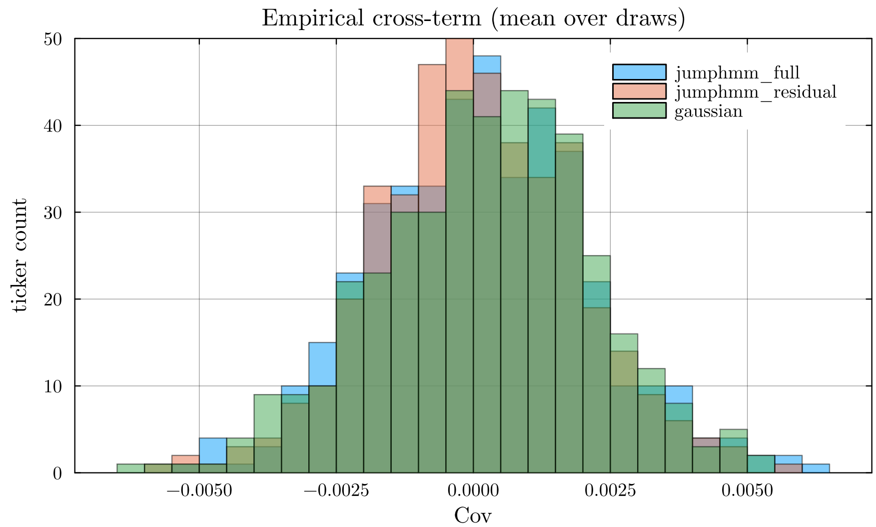

Figure S1: **Empirical normalized cross-covariance** $\mathrm{Cov}(\tilde{g}_i, g_m)/(\sigma_{\mathrm{gen},i}\,\sigma_m)$ **across the** 423**-ticker universe.** Histograms were stacked per generator, with each per-ticker value averaged over $R = 100$ generator growth-rate paths. The full-return JumpHMM, residual-fit JumpHMM, and Gaussian generators all clustered at zero, validating the independence assumption used to derive Eq. (7).

S1 Empirical cross-term diagnostic

The variance correction of Eq. (7) rests on the assumption $\mathrm{Cov}(\tilde{g}_i, g_m) \approx 0$, which is what allows the derivation to drop the cross-term in the expansion of $\mathrm{Var}(g_i)$. Because the per-asset JumpHMM marginals are fit to each asset's full growth-rate history, including its market-correlated component, the assumption is not automatic: a generator that picked up a residual market loading would invalidate the closed-form scaling. We tested the assumption empirically by sampling $R = 100$ generator growth-rate paths $\tilde{g}_i^{(r)}$ per ticker from each generator and computing the normalized cross-covariance against the real SPY path:

$$\mathrm{cov}_i^{\mathrm{norm}} = \frac{1}{R}\sum_{r=1}^{R} \frac{\mathrm{Cov}(\tilde{g}_i^{(r)},\, g_m)}{\sigma_{\mathrm{gen},i}\,\sigma_m}. \tag{S1}$$

Per-ticker values were plotted as histograms in Fig. S1. Across the 423 tickers, the full-return JumpHMM generator (used by the hybrid and naive composition methods), the residual-fit JumpHMM generator (used by the JumpHMM-on-residuals composition method), and the Gaussian baseline all clustered tightly around zero, with no per-generator mean bias outside sampling tolerance. The independence assumption was therefore satisfied for all three generators, and the derivation of Eq. (7) held in practice.

S2 Bias correction for daily-rebalanced backtests

This appendix derives the bias correction of Eq. (S14). Under daily rebalancing of the SIM-composed synthetic universe, the raw ensemble median annualized growth rate runs approximately 22 percentage points above the long-run prior. The variance correction of Eq. (7) is a precondition: it pins the synthetic per-asset second moments to their empirical targets, so the formula value of the diversification return on synthetic paths equals the formula value on real data. This 22-percentage-point difference is therefore not a structural consequence of the i.i.d.-residual draw at the second-order level; it comes from higher-order Taylor terms, calibration drift, and the documented attenuation of realized diversification returns [54], which the location-shift correction absorbs as a single empirical drag on log-wealth.

The synthetic paths reproduce per-asset marginals and single-factor cross-sectional dependence but discard temporal structure in the residual draw: at each trading day the per-asset residual $\varepsilon_i(t)$ is sampled independently from the previous day. Real equity residuals have small but non-negligible serial dependence at sub-monthly horizons [4], so the i.i.d.-residual generator family used here drops a real feature of the data. Consider a daily-rebalanced portfolio of N assets with constant target weights $w_i \geq 0$, $\sum_i w_i = 1$, reset to those weights at the start of each trading day. Throughout this derivation $g_i(t)$ is the annualized log-growth rate of Eq. (1), so the per-period log return on asset i is $g_i(t)\,\Delta t$ with $\Delta t = 1/252$ yr. The wealth process is:

$$W_t \;=\; W_{t-1} \sum_i w_i \exp\!\big(g_i(t)\,\Delta t\big), \tag{S2}$$

so the per-step log return on the portfolio is $\log(W_t/W_{t-1}) = \log\sum_i w_i \exp(g_i(t)\,\Delta t)$. We suppress the time argument, writing g_i for $g_i(t)$. A second-order Taylor expansion of the exponential about zero, $\exp(g_i\,\Delta t) = 1 + g_i\,\Delta t + \frac{1}{2}(g_i\,\Delta t)^2 + O((g\,\Delta t)^3)$, substituted into each summand of $\sum_i w_i \exp(g_i\,\Delta t)$ gives:

$$\begin{aligned}
\sum_i w_i \exp\!\big(g_i\,\Delta t\big) &= \sum_i w_i \big[\,1 + g_i\,\Delta t + \tfrac{1}{2}(g_i\,\Delta t)^2 + O\big((g\,\Delta t)^3\big)\big] && \text{(Taylor)}\\
&= \sum_i w_i + \Delta t \sum_i w_i\, g_i + \tfrac{1}{2}(\Delta t)^2 \sum_i w_i\, g_i^2 + O\big((g\,\Delta t)^3\big) && \text{(distribute)}\\
&= 1 + \Delta t \sum_i w_i\, g_i + \tfrac{1}{2}(\Delta t)^2 \sum_i w_i\, g_i^2 + O\big((g\,\Delta t)^3\big). && (\textstyle\sum_i w_i = 1)
\end{aligned} \tag{S3}$$

A second-order Taylor expansion of $\log(1+x)$ about $x = 0$ gives $\log(1+x) = x - \frac{1}{2}x^2 + O(x^3)$. Identifying $1 + x$ with the right-hand side of Eq. (S3), the order-Δt correction is $x := \Delta t \sum_i w_i\, g_i + \frac{1}{2}(\Delta t)^2 \sum_i w_i\, g_i^2$, and squaring keeps only the leading $(\Delta t)^2$ term: $x^2 = (\Delta t)^2(\sum_i w_i\, g_i)^2 + O((\Delta t)^3)$, since the cross term has an extra Δt factor. Substituting:

$$\begin{aligned}
\log\frac{W_t}{W_{t-1}} &= \log\Big(1 + x + O\big((g\,\Delta t)^3\big)\Big) && \text{(by Eq. (S3))}\\
&= x - \tfrac{1}{2}\,x^2 + O\big(x^3\big) && \text{(Taylor)}\\
&= \Delta t \sum_i w_i\, g_i + \tfrac{1}{2}(\Delta t)^2 \sum_i w_i\, g_i^2 - \tfrac{1}{2}(\Delta t)^2 \Big(\sum_i w_i\, g_i\Big)^2 + O\big((g\,\Delta t)^3\big) && \text{(plug in)}\\
&= \Delta t \sum_i w_i\, g_i + \tfrac{1}{2}(\Delta t)^2 \left(\sum_i w_i\, g_i^2 - \Big(\sum_i w_i\, g_i\Big)^2\right) + O\big((g\,\Delta t)^3\big). && \text{(group)}
\end{aligned} \tag{S4}$$

Dividing through by Δt extracts the portfolio's annualized one-step log-growth rate $g_p(t) := \Delta t^{-1} \log(W_t/W_{t-1})$, the portfolio counterpart of the per-asset $g_i(t)$ (still suppressing the time argument):

$$g_p \;=\; \sum_i w_i\, g_i \;+\; \tfrac{1}{2}\,\Delta t \left(\sum_i w_i\, g_i^2 \;-\; \Big(\sum_i w_i\, g_i\Big)^2\right) \;+\; O\big((\Delta t)^2\big). \tag{S5}$$

Restoring the time argument, we now take unconditional expectations of Eq. (S5) under stationarity. Define the per-asset annualized mean and variance:

$$\mu_i \;:=\; \mathbb{E}\big(g_i(t)\big), \qquad \sigma_i^2 \;:=\; \mathrm{Var}\big(g_i(t)\big), \tag{S6}$$

the per-rate covariance $\Sigma_{ij} := \mathrm{Cov}\big(g_i(t), g_j(t)\big)$, and the per-rate portfolio variance $\sigma_p^2 := \mathrm{Var}\big(\sum_i w_i g_i(t)\big) = w^\top \Sigma w$. The $O(\Delta t)$ correction in Eq. (S5) contains two squared quantities, $\sum_i w_i\, g_i(t)^2$ and $(\sum_i w_i\, g_i(t))^2$. Both expand by the second-moment identity, which holds for any random variable X with finite second moment:

$$\mathbb{E}\big(X^2\big) \;=\; \mathbb{E}\big(X\big)^2 \;+\; \mathrm{Var}(X). \tag{S7}$$

Applied to $X = g_i(t)$, with $\mathbb{E}(g_i) = \mu_i$ and $\mathrm{Var}(g_i) = \sigma_i^2$, then summed against w_i by linearity of expectation:

$$\begin{aligned}
\mathbb{E}\Big(\sum_i w_i\, g_i(t)^2\Big) \;&=\; \sum_i w_i\, \mathbb{E}\big(g_i(t)^2\big) && \text{(linearity)} \\
&=\; \sum_i w_i \big[\, \mu_i^2 + \sigma_i^2 \,\big] && \text{(by Eq. (S7))} \\
&=\; \sum_i w_i\, \mu_i^2 \;+\; \sum_i w_i\, \sigma_i^2. && \text{(S8)}
\end{aligned}$$

Applied to $X = \sum_i w_i g_i(t)$, with $\mathbb{E}(X) = \sum_i w_i\, \mu_i$ by linearity and $\mathrm{Var}(X) = w^\top \Sigma w = \sigma_p^2$ by definition of σ_p^2:

$$\begin{aligned}
\mathbb{E}\Big(\Big(\textstyle\sum_i w_i g_i(t)\Big)^2\Big) \;&=\; \mathbb{E}\Big(\textstyle\sum_i w_i g_i(t)\Big)^2 \;+\; \mathrm{Var}\Big(\textstyle\sum_i w_i g_i(t)\Big) && \text{(by Eq. (S7))} \\
&=\; \Big(\textstyle\sum_i w_i\, \mu_i\Big)^2 \;+\; \sigma_p^2. && \text{(S9)}
\end{aligned}$$

Substituting Eqs. (S8)–(S9) into the expectation of Eq. (S5) and grouping the σ_i^2 contributions separately from the μ_i contributions:

$$\mathbb{E}(g_p) \;=\; \sum_i w_i\, \mu_i + \underbrace{\tfrac{1}{2}\,\Delta t\Big(\sum_i w_i\, \sigma_i^2 - \sigma_p^2\Big)}_{D} + \tfrac{1}{2}\,\Delta t\Big(\sum_i w_i\, \mu_i^2 - \Big(\sum_i w_i\, \mu_i\Big)^2\Big) + O\big((\Delta t)^2\big), \tag{S10}$$

the Booth-Fama identity for the daily-rebalanced portfolio in annualized growth-rate units [46, 55]. The middle term, D, is the diversification return: the $O(\Delta t)$ correction equal to half the difference between the weighted-average per-asset variance $\sum_i w_i \sigma_i^2$ and the portfolio variance σ_p^2. The per-rate moments μ_i and σ_i^2 have units yr^{-1} and yr^{-2} respectively; the products $\Delta t\, \sigma_i^2$ and $\Delta t\, \sigma_p^2$ in Eq. (S10) are the annualized variances (yr^{-1}). The third term in Eq. (S10) is the cross-sectional dispersion of the per-asset annualized means, of order $\Delta t\, \mu^2 \sim 3 \times 10^{-5}\ \mathrm{yr}^{-1}$ at $\mu \sim 0.08\ \mathrm{yr}^{-1}$ and negligible against the variance term $D \sim \Delta t\, \sigma^2 \sim 4 \times 10^{-2}\ \mathrm{yr}^{-1}$; we drop it. The realized annualized compound growth rate $\bar{g}_p = (T\,\Delta t)^{-1} \log(W_T/W_0) = T^{-1} \sum_t g_p(t)$ converges to $\mathbb{E}(g_p)$ by the ergodic theorem under any stationary residual process, so the long-run limit is fixed by the one-time-slice unconditional joint distribution of $(g_i(t))_i$ and does not, in itself, depend on the serial structure of the residual draw.

To rewrite D in a form that makes its non-negativity manifest, expand $\sigma_p^2 = \sum_i w_i^2 \sigma_i^2 + 2 \sum_{i<j} w_i w_j \Sigma_{ij}$ and use the identity $\sum_i w_i(1 - w_i)\sigma_i^2 = \sum_{i<j} w_i w_j (\sigma_i^2 + \sigma_j^2)$ that holds for any weights summing to one:

$$D \;=\; \tfrac{1}{2}\,\Delta t\Big(\sum_i w_i \sigma_i^2 - \sigma_p^2\Big) \;=\; \tfrac{1}{2}\,\Delta t \sum_{i<j} w_i w_j\ \mathrm{Var}(g_i - g_j) \;\geq\; 0, \tag{S11}$$

so D equals one-half the annualized weighted-average pairwise growth-rate-difference variance and is therefore non-negative, with equality only when every pair (i, j) comoves perfectly ($\mathrm{Var}(g_i - g_j) = 0$) or all weight concentrates on a single asset. The rebalancing rule itself produces this effect: weights drift between rebalances, the rebalancer sells whatever has outperformed (its weight rose above target) and buys whatever has underperformed (its weight fell below target), and the trade systematically converts period-by-period cross-sectional dispersion into compounded growth. Markowitz [55] described this as the variance bonus of the long-run mean-variance allocation; Booth and Fama [46] called it the diversification return.

Substituting the single-index decomposition $g_i(t) = \alpha_i + \beta_i\, g_m(t) + \varepsilon_i(t)$, with market variance $\sigma_m^2 := \mathrm{Var}(g_m(t))$, per-asset residual variance $\sigma_{\varepsilon,i}^2 := \mathrm{Var}(\varepsilon_i(t))$, and $\mathrm{Cov}(g_m, \varepsilon_i) = 0$ (verified in Sec. S1), into Eq. (S11) gives:

$$\mathrm{Var}(g_i - g_j) \;=\; (\beta_i - \beta_j)^2 \sigma_m^2 \;+\; \mathrm{Var}(\varepsilon_i - \varepsilon_j), \tag{S12}$$

Table S1: **Descriptive statistics and stylized-facts tests for SPY daily excess growth rates.** In-sample (IS): 2014–2024 ($T = 2{,}766$); out-of-sample (OoS): 2025 ($T = 249$). Gaussian and Laplace columns show maximum likelihood estimation (MLE) fits to the in-sample data. JB: Jarque-Bera normality test. LB: Ljung-Box autocorrelation test at lag 20. * denotes rejection at $\alpha = 0.05$.

Statistic	**IS Observed**	**OoS Observed**	**Gaussian**	**Laplace**
Mean (annualized, %)	6.31	11.60	6.31	14.19
Std Dev (annualized, %)	214.50	220.62	214.46	205.93
Skewness	-0.753	-1.093	0	0
Excess Kurtosis	7.715	6.867	0	3
JB normality test	reject* ($p < 0.001$)	reject* ($p < 0.001$)	n/a	n/a
LB test on G_t (lag 20)	reject* ($p < 0.001$)	reject* ($p = 0.019$)	n/a	n/a
LB test on $\lvert G_t \rvert$ (lag 20)	reject* ($p < 0.001$)	reject* ($p < 0.001$)	n/a	n/a

since g_m and the residuals are uncorrelated, so the diversification return splits into a market-beta-dispersion component and an idiosyncratic component:

$$D \;=\; \underbrace{\tfrac{1}{2}\,\Delta t\,\sigma_m^2 \sum_{i<j} w_i w_j (\beta_i - \beta_j)^2}_{D_{\mathrm{mkt}}} \;+\; \underbrace{\tfrac{1}{2}\,\Delta t \sum_{i<j} w_i w_j \;\mathrm{Var}(\varepsilon_i - \varepsilon_j)}_{D_\varepsilon}, \tag{S13}$$

where the market-beta-dispersion component D_{mkt} equals $\frac{1}{2}\,\Delta t\,\sigma_m^2[\sum_i w_i\beta_i^2 - (\sum_i w_i\beta_i)^2]$, the weighted cross-sectional β dispersion scaled by the annualized market variance $\Delta t\,\sigma_m^2$. Across the 423-ticker universe the empirical β distribution spans $[0.01, 1.79]$ with interquartile range $[0.81, 1.18]$ and the SPY annualized variance $\Delta t\,\sigma_m^2 \approx 0.04\ \mathrm{yr}^{-1}$, so D_{mkt} contributes at most a few percentage points per year for any diversified equity portfolio. The idiosyncratic component D_ε is the dominant term: with residuals approximately orthogonal across assets at the same time slice, $\mathrm{Var}(\varepsilon_i - \varepsilon_j) \approx \sigma_{\varepsilon,i}^2 + \sigma_{\varepsilon,j}^2$, and D_ε scales with the weighted-average annualized idiosyncratic variance $\Delta t\,\sigma_{\varepsilon,i}^2$ per name. Concentrated allocators on high-idiosyncratic-volatility names therefore have large formula values of D_ε.

The i.i.d. construction of the residual draw fixes both terms of Eq. (S13). The variance correction of Eq. (7) pins the synthetic per-asset variance $\sigma_{\mathrm{gen},i}^2 = \beta_i^2\sigma_m^2 + \sigma_{\varepsilon,i}^2$ to its empirical target, so the formula value $D = D_{\mathrm{mkt}} + D_\varepsilon$ on synthetic paths equals the formula value under any alternative residual generator matched on the same unconditional second moments, including a stationary AR(1) residual generator $\varepsilon_i(t) = \phi\,\varepsilon_i(t-1) + \eta_i(t)$ with i.i.d. innovations η_i tuned so that $\mathrm{Var}(\varepsilon_i) = \sigma_{\eta,i}^2/(1-\phi^2) = \sigma_{\varepsilon,i}^2$. The AR(1) parameter ϕ leaves the unconditional joint distribution of $(g_i(t))_i$ at a single time slice unchanged, which by Eqs. (S10), (S13) means ϕ enters at zeroth order in $\mathbb{E}(g_p)$ and at first order only in the long-run variance of the sample mean ($\mathrm{Var}(\bar{g}_p) \to T^{-1}\,\sigma_p^2\,(1+\phi)/(1-\phi)$). The synthetic-ensemble median across 1,000 paths therefore converges to the same formula value whether the residuals are drawn i.i.d. or with positive serial autocorrelation: the i.i.d. versus AR(1) distinction shows up in per-path sampling noise, not in the ensemble median itself. The raw median's $\sim$22-percentage-point excess over an external long-run growth prior is therefore not caused by i.i.d. sampling *across time* at the level of the second-order formula; the AR(1)-versus-i.i.d. serial structure does not move it. The more direct structural contributor is contemporaneous: the split above assumes the residuals are orthogonal across assets at a given time slice, which holds for the per-asset i.i.d. generator but not for real residuals, where common sector and style factors correlate the ε_i and shrink the realized $\mathrm{Var}(\varepsilon_i - \varepsilon_j)$ below the $\sigma_{\varepsilon,i}^2 + \sigma_{\varepsilon,j}^2$ that the independent generator produces. Matching each asset's marginal variance does not restore that cross-asset covariance, so the synthetic D_ε is expected to exceed its real-market counterpart when the omitted residual covariance is positive; quantifying that contribution requires a joint-residual experiment we did not run. The residual excess beyond this contemporaneous term is the combined empirical magnitude of the higher-moment corrections to the Taylor expansion (heavy-tailed JumpHMM marginals have non-trivial third and fourth cumulants), the time-varying volatility and correlation across the 2014–2024 calibration window relative to a long-run prior, and the documented attenuation of realized diversification returns on real equity portfolios at all rebalancing horizons [54]. The location-shift correction of Eq. (S14) treats this composite shortfall as a single empirical drag b to be absorbed and is agnostic to its decomposition.

Table S2: **Diversification-return decomposition on a 20-asset target-weight allocator deployed against the hybrid-composed 2025 universe.** Median path-mean annualized growth rate $\bar{g}_p$ across the synthetic ensemble. Equal-weight buy-and-hold serves as a passive baseline; the target-weight allocation is reported under static (no intra-year rebalance) and daily-rebalanced regimes, with and without a signal on top of the target weights. The static-to-daily increase of $+27.73$ pp on the same target weights isolates the diversification return of Eq. (S13). The same allocator deployed against the real 2025 growth-rate history is shown for comparison.

Allocation rule (synthetic 2025 universe)	$\bar{g}_p$ **(%/yr)**
Equal-weight buy-and-hold	7.68
Target-weight, static (no rebalance)	1.97
Target-weight, daily rebalanced	29.69
Target-weight, daily rebalanced + signal	29.83
Real 2025 history, same allocator	11.50

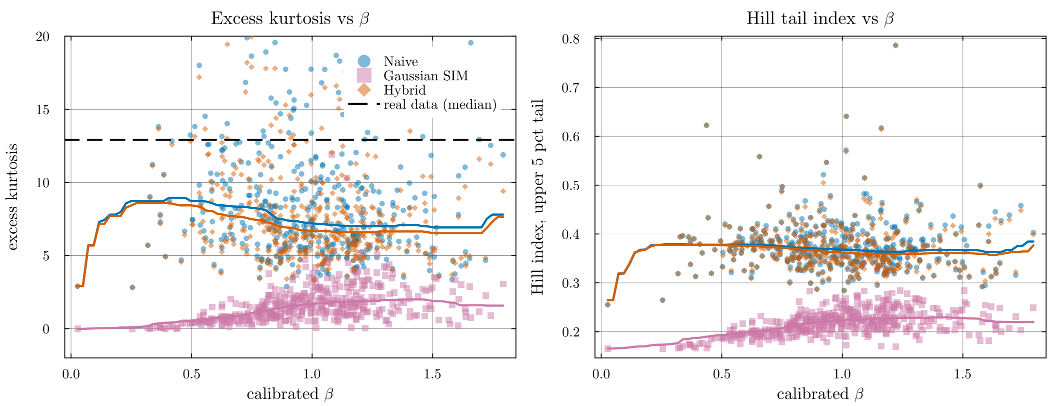

Figure S2: **Heavy-tail preservation: excess kurtosis and Hill tail index vs calibrated β.** Each point is a per-ticker median over 100 replications. *Left:* excess kurtosis, y-axis clipped to $[-2, 20]$ for readability. The dashed line marks the median real-data excess kurtosis across the universe ($\kappa_{\text{real}} \approx 13$); the hybrid and naive composition methods (orange and blue) track the generator's kurtosis, while Gaussian SIM collapses to $\kappa \approx 1$. *Right:* Hill tail-index estimator on the upper 5% of $|g|$. The hybrid and naive medians remain near 0.37 throughout the β range; Gaussian SIM drops to ~ 0.22, indicating the shift toward a thinner-tailed distribution. The hybrid's median being below κ_{real} reflects the JumpHMM marginal's own kurtosis rather than a composition-rule distortion: hybrid tracks naive (both retain the generator's tails) while Gaussian SIM does not.

To measure the scale of the bias, we deployed a 20-asset long-only target-weight allocator (calibration window 2014–2024, deployment year 2025, names with calibrated (α, β) and $\max \beta = 1.21$, 5 basis-point round-trip cost) against the hybrid-composed universe of the main text and decomposed the realized path-mean annualized growth rate $\bar{g}_p \equiv (T\,\Delta t)^{-1} \log(W_{p,T}/W_{p,0})$ into static and rebalancing-driven components (Table S2). The static-to-daily-rebalance switch on the same target weights produced a $+27.73$ percentage-point increase that isolates the diversification-return term of Eq. (S10); adding a signal on top of the daily rebalance moved the median by a further $+0.13$ pp, negligible next to the rebalancing-driven diversification return. The same allocator on the real 2025 growth-rate history produced $11.5\%/\text{yr}$, leaving the synthetic median running ~ 22 percentage points above reality. The decomposition is consistent with a diversification-return explanation rather than a calibration error: the allocator choice, the cost model, and the marginal calibration match the real 2025 deployment, and the residual draws are independent across tickers, omitting the contemporaneous cross-asset covariance real markets exhibit. Confirming that this omission accounts for the full ~ 22 percentage-point gap would require an ablation that reintroduces realistic cross-asset residual covariance while holding the marginals and allocator fixed.

The correction acts in log-wealth space, subtracting a constant drift from every path so that the ensemble median annualized growth rate matches an external long-run prior $\bar{g}^{\text{prior}}$. At any fixed step the transform is monotone in $W_{p,t}$, so it leaves the cross-path rank order by terminal wealth intact;

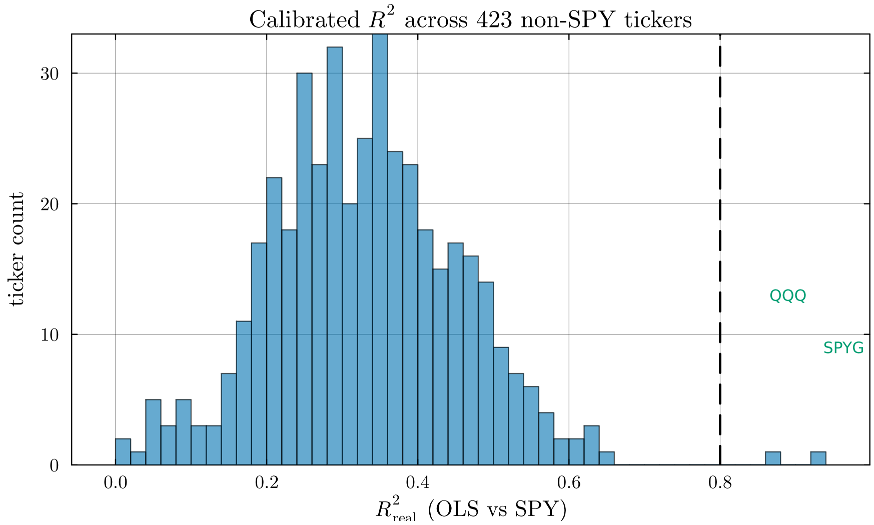

Figure S3: **Calibrated R^2 distribution across the 423 non-SPY tickers.** Histogram of OLS R^2 against SPY over the 2014–2024 window. The two tickers crossing the $R^2_{\text{preserve}} = 0.80$ threshold (dashed line) are index ETFs (QQQ, SPYG); the highest-R^2 individual stock (BLK, 0.645) falls ~ 0.16 below the threshold. The R^2-preserving branch's $n = 2$ occupancy is therefore a property of the single-factor SIM on S&P 500 names against SPY, not a sample-size artifact or a data-peeked threshold.

because it shifts every log-return increment by the same constant, it also leaves per-path increment variances (the volatility envelope) unchanged. Each path-p wealth trajectory $W_{p,t}$ at integer step t is multiplied by a uniform exponential drag at rate b, set to close the difference between the raw median annualized growth rate and the prior $\bar{g}^{\text{prior}}$:

$$W^{\text{corr}}_{p,t} \;=\; W_{p,t} \cdot \exp\bigl(-\, b \cdot t\,\Delta t\bigr), \qquad b \;=\; \operatorname*{median}_{p} \bar{g}_p \;-\; \bar{g}^{\text{prior}}, \tag{S14}$$

with Δt the integration step in years and b, $\bar{g}_p$, $\bar{g}^{\text{prior}}$ all in the same per-year growth-rate units. Working in log-space keeps the correction multiplicative on wealth, the natural scale for compounding returns. Because the drift subtracted per unit time is constant, the correction is a linear detrend in log-wealth: it does shift drawdown dates and depths within a path, so it is a level calibration rather than a drawdown-preserving transform. What it preserves is the cross-path terminal-wealth ordering and the per-path volatility envelope, the properties later ranking and risk diagnostics rely on. The prior $\bar{g}^{\text{prior}}$ is set from external long-run data and is typically near $8\%/\text{yr}$ at the time of writing: the prevailing risk-free rate (roughly $4.5\%/\text{yr}$ on a five-year Treasury basis) plus the historical long-run equity premium of 4–6 pp [56, 57], less 1–2 pp for fees, turnover, and market-impact frictions, leaving ~ 3.5 pp of active premium net of costs. Using a documented prior instead of the realized test-year outcome avoids circular calibration: matching the realized year would force the synthetic median to collapse onto the test sample by construction and would misrepresent typical years. In our deployment, the realized 2025 was an above-typical equity year, so matching its $11.5\%/\text{yr}$ outcome would have suppressed the corrected ensemble below its appropriate long-run level. As a check, the corrected ensemble matched the prior at its median by construction, and the real-2025 outcome landed at the 63rd percentile of the corrected ensemble, consistent with 2025 being an above-typical equity year.

Several alternative fixes were considered and rejected. A path-filtering rule that drops synthetic paths above a $\bar{g}_p$ threshold turns the threshold itself into a tuning target and removes only a subset of the bias's consequences while leaving the underlying bias intact. Inflating per-trade transaction costs to absorb the diversification return would require round-trip costs of ~ 70 basis points versus the realistic 1–10 basis points seen on liquid US equity, shifting the misrepresentation from the synthetic generator to its cost model. Drifting the calibrated parameters between calibration and deployment to widen the predictive envelope, with drift scale varied across $\{0, 1, 2, 3, 5\}$ multiples of the OLS standard error, moved the median Sharpe ratio by less than 0.01 across the sweep, confirming that β is estimated too tightly for standard-error-scale drift to absorb the bias. Imposing an AR(1) on the residual draw inside the existing construction would conflict with the copula reorder step, which destroys temporal structure by construction; recovering the AR(1) requires a block-ordered copula sampler, which is a structural redesign rather than a correction applied to already-composed paths. The block bootstrap of empirical residuals [52], applied per ticker as here, preserves each ticker's own temporal residual structure but not the contemporaneous cross-asset covariance (a joint or aligned bootstrap, or a residual copula, would be required for that); it is a valid alternative generator, listed as a future-work item that replaces the JumpHMM marginal rather than correcting it. A multiplicative haircut on the raw-versus-static deviation, $W^{\mathrm{corr}} = W_{\mathrm{static}} \cdot (W_{\mathrm{raw}}/W_{\mathrm{static}})^k$, hits the same target at k chosen to match the prior, but its per-path rank correlation with the raw path drops well below one, so unlike the location shift it reorders which paths are best and worst; it also inherits the static allocation's drawdown pattern, with drawdowns widening materially relative to the raw path.

The location-shift correction of Eq. (S14), by contrast, preserves the per-path rank order and the volatility envelope, so the corrected ensemble still supports the ensemble-shape diagnostics the paper validates: percentile overlays of real outcomes against the synthetic distribution, tail-shape analysis, and the relative ranking of strategies evaluated on the same corrected paths. The drawdown distribution does shift under the correction, as under any level recalibration, so drawdown-based diagnostics are read on the uncorrected ensemble. It does not remove the underlying i.i.d.-residual limitation, so the residual autocorrelation structure itself, absolute Sharpe under daily rebalancing, and the choice of rebalancing frequency remain unaddressed by the correction; the structural fix for those targets requires redesigning the AR(1) or GARCH residual structure around the copula reorder step, listed as future work.

S3 Validation metric definitions

The distributional and tail metrics reported in the main results were defined as follows. The two-sample Kolmogorov-Smirnov statistic between the empirical cumulative distribution functions (CDFs) of the observed and simulated growth-rate samples (G_{obs} of length n and G_{sim} of length m) is $D = \sup_x |F_n(x) - F_m(x)|$ [43, 44]; the test rejects when D exceeds the critical value at level $\alpha = 0.05$. The Anderson-Darling statistic places greater weight on the tails via the integral $A^2 = nm/(n+m) \int [F_n(x) - F_m(x)]^2/[F(x)(1 - F(x))]\, dF(x)$, where F is the pooled empirical CDF [45]. The Wasserstein-1 distance between two empirical distributions of equal length is the average of sorted-quantile differences:

$$W_1 = \frac{1}{n} \sum_{i=1}^{n} \left|G_{\mathrm{obs}}^{(i)} - G_{\mathrm{sim}}^{(i)}\right|, \tag{S15}$$

with $G^{(i)}$ the ith order statistic. The Hill upper-tail index is the classical order-statistic estimator:

$$\hat{\xi} = \frac{1}{k-1} \sum_{i=1}^{k-1} \log\left(\frac{X_{(i)}}{X_{(k)}}\right), \tag{S16}$$

with $X_{(1)} \geq X_{(2)} \geq \ldots$ the ordered upper-tail absolute returns; k is set to the upper 5% of the sample. For a Pareto-tailed distribution with shape α, $\hat{\xi} \approx 1/\alpha$. The Hellinger distance between two density estimates p and q on a common bin grid is:

$$H(p, q) = \frac{1}{\sqrt{2}} \left\|\sqrt{p} - \sqrt{q}\right\|_2 \;\in\; [0, 1], \tag{S17}$$

with $H = 0$ for identical densities and $H = 1$ for disjoint supports.

S4 Transition matrix internals and partition diagnostics

We inspected the three internal components of the fitted HMM-WJ on SPY with $N = 100$ states (Fig. S5). The fitted Laplace cumulative distribution function overlays the empirical CDF closely, with the 99 vertical lines marking the equal-probability quantile boundaries Q_k that define the hidden states, confirming the Laplace fit as an accurate marginal model for the SPY excess growth-rate series and validating the partition rule of Eq. (1). The estimated transition matrix $\mathbf{T}$, plotted in $\log_{10}$ color scale, shows a dominant near-diagonal band that reflects short-range regime persistence, with most off-diagonal mass concentrated within ± 5 states of the diagonal. The expected natural residence

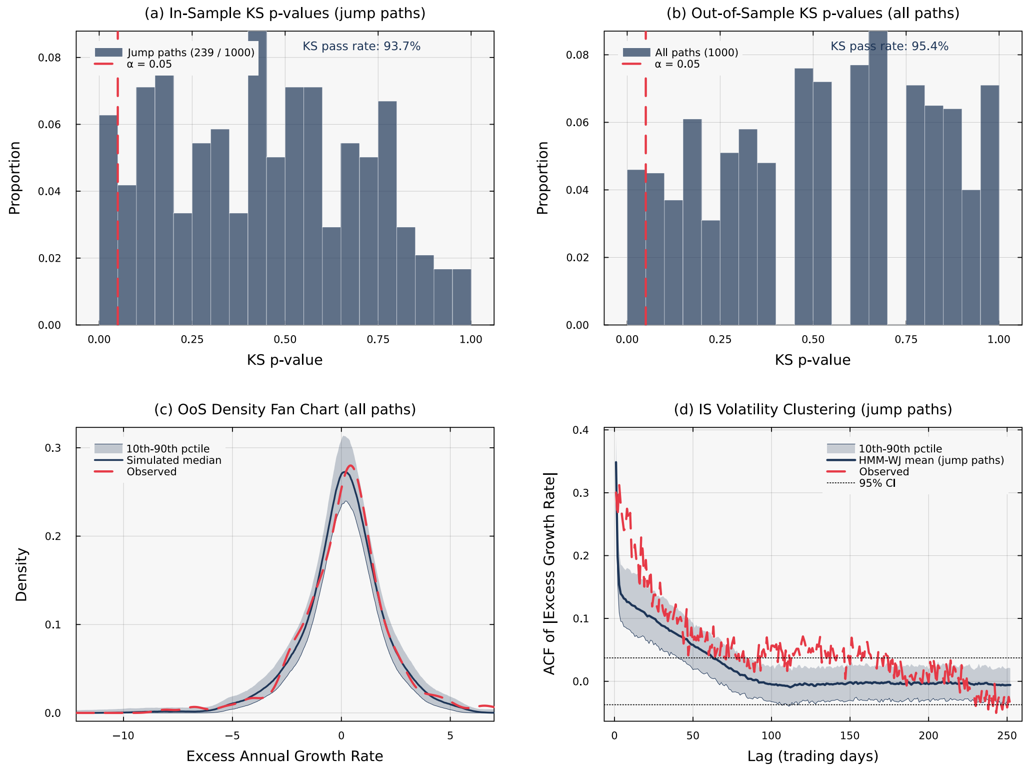

Figure S4: **Statistical validation of HMM-WJ** ($N = 100$, $\alpha = 0.05$). Panels (a) and (d) condition on the $\sim 24\%$ of in-sample paths containing jump events. (a) In-sample KS p-values for 239 jump paths (pass rate 93.7%). (b) Out-of-sample KS p-values for all 1,000 paths (pass rate 95.4%). (c) Out-of-sample density fan chart; the observed density falls within the simulation envelope. (d) In-sample ACF of $|G_t|$ for jump paths; the mean closely tracks the observed ACF.

time $1/(1 - T_{kk})$ for each state, plotted on a $\log_{10}$ scale, with the bottom tail set $\mathcal{S}_-$ (states 1–5, red) and the top tail set $\mathcal{S}_+$ (states 96–100, teal) highlighted, shows how far these residence times fall short of the empirical clustering scale. Under pure Markovian dynamics every state, including the tail states, exits within 1–2 steps on average, far too quickly to reproduce empirical volatility clustering; the dashed line marks the fitted mean jump duration $\lambda = 100$ steps and quantifies the two-order-of-magnitude gap between natural residence times and the jump-duration override. Quantile-bin occupancy was approximately uniform over the in-sample window, consistent with the Laplace fit's calibration to the empirical CDF; the residual sample-size variation across bins was smaller than the within-bin emission variance, so all 100 states received adequate statistical support.

S5 Emission distribution and semi-Markov comparison

To isolate the contributions of the emission family and the duration model, we compared three variants on SPY (Table S3): the HSMM baseline of Bulla and Bulla [16] ($K = 8$ states, negative-binomial dwell times, Student-t emissions), HMM-WJ with the original Gaussian emissions ($N = 100$), and HMM-WJ with Student-t emissions ($N = 100$). The HMM-WJ variants exceeded the HSMM on in-sample distributional fidelity by 15 percentage points on KS and 49 percentage points on AD regardless of emission choice; we attribute this difference to the fine-grained quantile partition over the coarse $K = 8$ HSMM partition. Holding the HMM-WJ machinery fixed, the Student-t ($\nu = 5$) upgrade over the Normal baseline closed the in-sample excess-kurtosis gap from a 5.5 value under Normal emissions to a 7.6 value under Student-t emissions, against an observed 7.7 value, while leaving KS, AD, ACF-MAE, Wasserstein-1, and Hellinger essentially unchanged. The out-of-sample column shows the same ordering on tail moments and a narrower spread on distributional pass rates, consistent with the single-year ($T = 249$) sample.

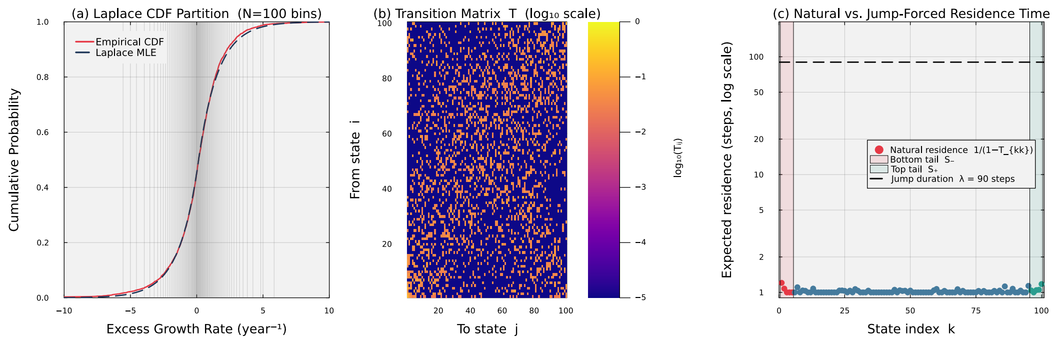

Figure S5: **Fitted HMM-WJ internals for SPY with $N = 100$ states.** Panel (a) overlays the fitted Laplace CDF on the empirical CDF; the 99 vertical lines mark the equal-probability quantile boundaries Q_k that define the hidden states. Panel (b) displays the estimated transition matrix $\mathbf{T}$ in $\log_{10}$ color scale; the dominant near-diagonal band reflects short-range regime persistence. Panel (c) plots the expected natural residence time $1/(1 - T_{kk})$ for each state on a $\log_{10}$ scale, with the bottom tail set $\mathcal{S}_-$ (states 1–5, red) and the top tail set $\mathcal{S}_+$ (states 96–100, teal) highlighted. The dashed horizontal line marks the fitted mean jump duration $\lambda = 100$ steps, quantifying the two-order-of-magnitude gap between natural residence times and the jump-duration override.

Table S3: **Three-way comparison isolating the emission distribution and the duration model for SPY.** Standard errors in parentheses; bold marks the best value per row. HSMM: hidden semi-Markov model ($K = 8$ states, negative-binomial dwell times, Student-t emissions) [16]; HMM-WJ: hybrid hidden Markov model with jump-duration override ($N = 100$). All entries are computed from 1,000 simulated paths at significance level $\alpha = 0.05$.

Metric	**HSMM (Student-t)**	**HMM-WJ (Gaussian)**	**HMM-WJ (Student-t)**
States	$K = 8$	$N = 100$	$N = 100$
Parameters	18	4	4
In-sample (2,766 trading days, 2014–2024)			
KS pass rate (%)	82.0 (1.2)	97.0 (0.5)	**97.6** (0.5)
AD pass rate (%)	42.5 (1.6)	90.5 (0.9)	**91.3** (0.9)
Excess kurtosis	4.8 (0.08)	5.5 (0.03)	**7.6** (0.14)
ACF-MAE	0.059 ($<$0.001)	**0.052** ($<$0.001)	**0.052** ($<$0.001)
Wasserstein-1	0.176 (0.001)	**0.101** (0.002)	**0.101** (0.001)
Hellinger	0.113 ($<$0.001)	0.076 ($<$0.001)	**0.075** ($<$0.001)
Out-of-sample (249 trading days, 2025)			
KS pass rate (%)	**96.2** (0.6)	95.0 (0.7)	94.4 (0.7)
AD pass rate (%)	**96.7** (0.6)	95.9 (0.6)	95.1 (0.7)
Excess kurtosis	4.1 (0.13)	4.9 (0.08)	**6.1** (0.13)
ACF-MAE	0.042 ($<$0.001)	0.040 ($<$0.001)	**0.039** ($<$0.001)
Wasserstein-1	0.287 (0.002)	**0.275** (0.005)	0.282 (0.006)
Hellinger	0.239 (0.001)	0.212 (0.001)	**0.210** (0.001)

Table S4: **Hybrid composition method broken out by branch.** N_{tickers}: number of tickers assigned to each branch by the $R^2_{\text{preserve}} = 0.80$ threshold. The clipped variance-preserving branch does not trigger anywhere in the universe ($\rho_i < 0.90$ for all tickers) and is omitted. The two trackers in the R^2-preserving branch (QQQ and SPYG) attain a KS pass rate of 99%, consistent with the branch recovering both β and R^2 by construction.

Branch	N_{tickers}	$\lvert\hat{\beta} - \beta\rvert$	$\lvert\hat{R}^2 - R^2_{\text{real}}\rvert$	KS pass (%)	W_1	κ
`hybrid`	421	0.018	0.021	50.1	0.273	7.142
`r2-preserve`	2	0.005	0.001	99.0	0.109	11.525

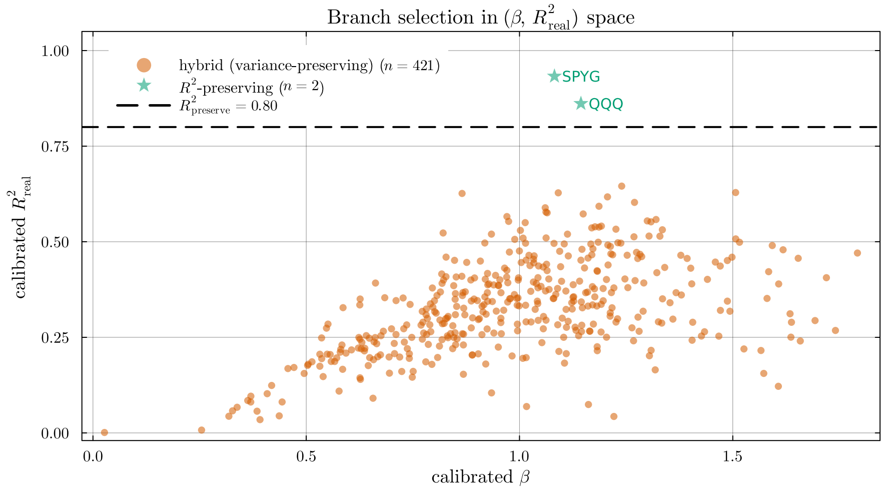

Figure S6: **Branch assignment in calibrated $(\beta, R^2_{\text{real}})$ space.** The $R^2_{\text{preserve}} = 0.80$ threshold (dashed line) separates tracker assets from single stocks. In this universe, only QQQ and SPYG exceed the threshold, and both are assigned to the R^2-preserving branch. The remaining 421 tickers populate the variance-preserving branch; the clipped variance-preserving branch is unused because the market loading ratio $\rho_i = \beta_i^2 \sigma_m^2 / \sigma_{\text{gen},i}^2$ stays below 0.90 for every ticker.

S6 State-resolution sensitivity

We swept the partition resolution over $N \in \{30, 60, 90, 100, 150, 200, 350\}$ and re-evaluated the in-sample KS and AD pass rates for HMM-NJ and HMM-WJ. Pass rates were flat across $N \in \{60, \ldots, 200\}$ with sub-percent variation, indicating that the model was insensitive to partition resolution in this range. At $N = 30$ both models lost approximately 5 percentage points on the AD test, reflecting the coarser quantile bins' reduced ability to track tail behavior. At $N = 350$ several quantile bins were never visited in the historical record, leaving the corresponding emission parameters undefined; this set the practical upper bound on N for the SPY in-sample window. We adopted $N = 100$ for all subsequent experiments.

S7 Cross-asset generalization

To verify that the hybrid HMM generalized beyond SPY, we re-ran the full fit-and-validate procedure on three single-stock tickers chosen to span sector and volatility regimes: NVDA (semiconductors, high volatility), JNJ (consumer staples, low volatility), and JPM (financials, moderate volatility). Per-ticker tune outputs at $(n_{\text{paths}} = 200, \text{seed} = 1234)$ are reported in Table S7. Two qualifications apply: the tune objective combines an ACF target and a kurtosis target through a fixed-weight sum, and the kurtosis penalty has substantial sample variance on heavy-tailed paths, so the specific argmin location is sensitive to Monte-Carlo noise at finite n_{paths}; the (ϵ^*, λ^*) values in the table are reproducible at the stated configuration but should not be over-interpreted as canonical per-ticker optima. The quantities the tune procedure produces reliably are the resulting distributional metrics at the chosen parameter values: in-sample HMM-WJ KS pass rates exceeded 98.9% on all three tickers, indicating a close fit to each ticker's training-window distribution. Out-of-sample generalization on the 249-day 2025 hold-out varied by asset: NVDA retained 97.7% KS pass, comparable to its in-sample fit, while JNJ and JPM attenuated to 74.0% and 79.2% respectively, reflecting larger between-window shifts in the per-ticker marginal distribution than the SPY series exhibited.

S8 SIM composition derivations and properties verification

The variance-corrected single-index composition states two closed-form results, the variance scale of Eq. (7) and the residual-variance target of Eq. (10), and asserts that the composed series g_i satisfies the three preservation criteria (SIM recovery, marginal fidelity, cross-sectional dependence) in the

Table S5: **Aggregate metrics by β quartile, across composition methods.** Quartile cutoffs are computed from the calibrated β distribution across the 423-ticker universe. The hybrid KS pass rate declines gently from 67% in Q1 to 44% in Q4 as the market loading ratio ρ_i grows, consistent with the tail-attenuation caveat noted alongside Algorithm 2; the naive KS pass rate falls from 16% to under 1% over the same range.

β bucket	Composer	$\lvert\hat{\beta} - \beta\rvert$	KS pass (%)	W_1	κ	Var(g)
Q1 (low β)	Naive	0.018	15.7	0.414	8.971	18.1
	Gaussian SIM	0.015	0.0	0.620	0.622	14.0
	Hybrid	0.016	67.3	0.200	8.369	14.6
	JumpHMM-on-residuals	0.015	73.2	0.204	7.059	14.3
	Block bootstrap	0.015	71.3	0.208	6.286	13.9
	GARCH(1,1)-t	0.015	57.8	0.253	5.255	13.4
Q2	Naive	0.021	0.9	0.632	7.510	25.1
	Gaussian SIM	0.016	0.2	0.653	1.509	18.0
	Hybrid	0.017	45.9	0.254	6.889	19.0
	JumpHMM-on-residuals	0.017	82.7	0.200	7.799	18.7
	Block bootstrap	0.016	79.8	0.197	7.213	18.0
	GARCH(1,1)-t	0.015	76.4	0.227	6.470	17.8
Q3	Naive	0.023	0.4	0.802	7.390	33.3
	Gaussian SIM	0.017	1.4	0.688	1.947	23.4
	Hybrid	0.018	44.6	0.297	6.904	24.4
	JumpHMM-on-residuals	0.018	78.3	0.242	7.454	24.0
	Block bootstrap	0.017	75.0	0.236	6.888	22.8
	GARCH(1,1)-t	0.016	71.0	0.263	6.265	21.9
Q4 (high β)	Naive	0.029	0.8	0.976	7.233	51.2
	Gaussian SIM	0.022	0.6	0.825	2.054	36.3
	Hybrid	0.022	43.7	0.363	6.739	37.1
	JumpHMM-on-residuals	0.022	69.1	0.313	6.842	37.0
	Block bootstrap	0.021	67.2	0.311	6.334	35.3
	GARCH(1,1)-t	0.021	64.5	0.361	6.064	34.7

Table S6: **Clipping-branch stress test under market-volatility scaling.** The market growth-rate path was rescaled by $\gamma \in \{1, 2, 3\}$ with the calibrated marginals held fixed, and the full hybrid composition method re-evaluated on the 423-ticker universe. As γ rises, the share of clipped tickers grows and the hybrid KS pass rate increases rather than falls.

γ	**Clipped (%)**	**Hybrid KS (%)**
1	0.0	50.3
2	76.4	71.8
3	95.2	77.3

Table S7: **Cross-asset generalization of HMM-WJ across three single-stock tickers spanning sector and volatility regimes.** Per-ticker tune outputs (ϵ^*, λ^*) at $n_{\text{paths}} = 200$ and seed $= 1234$, and the corresponding HMM-WJ pass rates on 1,000 simulated paths at $\alpha = 0.05$ against each ticker's in-sample (2014–2024) and out-of-sample (2025) growth-rate history. Standard errors in parentheses (binomial SE for pass rates). The (ϵ^*, λ^*) values are reproducible at the stated tune configuration but the composite objective has multiple near-equivalent local minima, so they are reported as the parameter values used for the IS / OoS fit metrics rather than as canonical per-ticker optima.

Ticker	ϵ^*	λ^*	**IS KS (%)**	**IS AD (%)**	**OoS KS (%)**	**OoS AD (%)**
NVDA (semiconductors)	10^{-4}	70	98.9 (0.3)	96.7 (0.6)	97.7 (0.5)	96.1 (0.6)
JNJ (consumer staples)	10^{-4}	70	99.4 (0.2)	96.3 (0.6)	74.0 (1.4)	61.3 (1.5)
JPM (financials)	10^{-4}	80	99.1 (0.3)	95.7 (0.6)	79.2 (1.3)	77.7 (1.3)

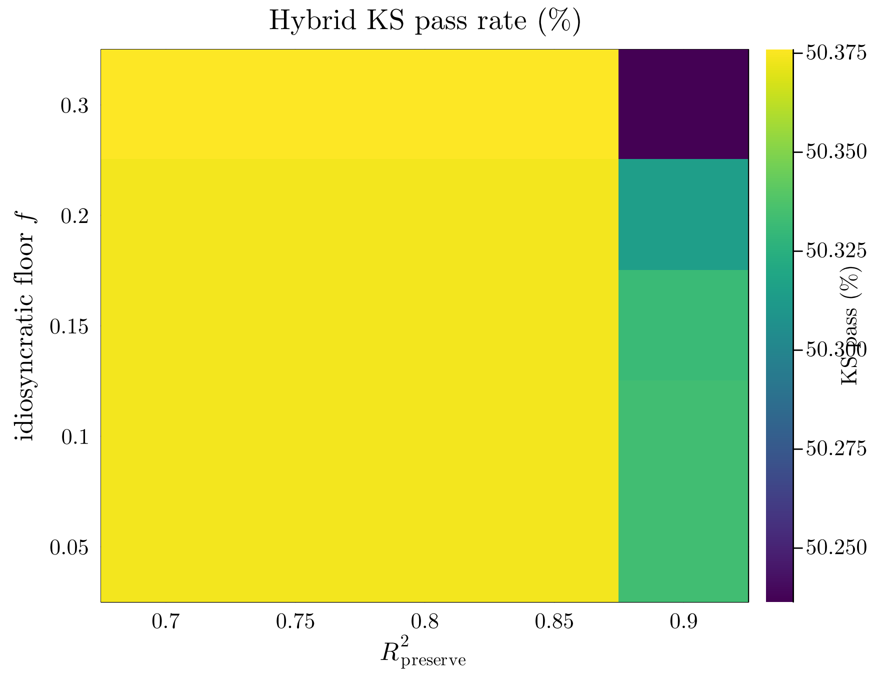

Figure S7: **Hybrid KS pass rate across a 5×5 hyperparameter grid.** Columns index the R^2-preserve threshold $R^2_{\text{preserve}} \in \{0.70, 0.75, 0.80, 0.85, 0.90\}$; rows index the idiosyncratic floor $f \in \{0.05, 0.10, 0.15, 0.20, 0.30\}$. Hybrid KS pass rate varies from 50.24% to 50.38% across the 25 cells, a spread of 0.14 percentage points. The parameter values $(0.80, 0.10)$ used throughout the paper are indistinguishable from every other grid point within sampling noise, indicating that the main result is not a data-peeked local optimum.

Table S8: **Seed-uncertainty summary across 8 random seeds.** Each cell reports mean $\pm$ standard deviation (SD) across the 8 seeds in $\{1234, 2345, 3456, 4567, 5678, 6789, 7890, 8910\}$, with each seed running the full 423-ticker $\times$ 100-replication protocol. Conditional-variance paths for GARCH(1,1)-t are re-simulated at scoring time so its seed SD is informative. The between-composition-method differences in KS pass rate exceed the seed-induced SD by roughly two orders of magnitude (hybrid SD 0.18 pp vs 25 pp gap to JumpHMM-on-residuals).

Composer	$\lvert\hat{\beta} - \beta\rvert$	$\lvert\hat{R}^2 - R^2_{\text{real}}\rvert$	KS pass (%)	AD pass (%)	κ	Hill
Naive	0.022 ± 0.000	0.088 ± 0.000	4.41 ± 0.07	2.71 ± 0.02	7.746 ± 0.015	0.369 ± 0.000
Gaussian SIM	0.017 ± 0.000	0.008 ± 0.000	0.60 ± 0.04	0.50 ± 0.01	1.428 ± 0.005	0.217 ± 0.000
Hybrid	0.018 ± 0.000	0.021 ± 0.000	50.40 ± 0.18	47.50 ± 0.19	7.150 ± 0.011	0.364 ± 0.000
JumpHMM-on-residuals	0.018 ± 0.000	0.015 ± 0.000	75.61 ± 0.11	70.51 ± 0.12	7.250 ± 0.013	0.365 ± 0.000
Block bootstrap	0.017 ± 0.000	0.020 ± 0.000	73.24 ± 0.09	66.79 ± 0.11	6.676 ± 0.011	0.330 ± 0.000
GARCH(1,1)-t	0.017 ± 0.000	0.036 ± 0.000	66.96 ± 0.30	61.00 ± 0.26	6.072 ± 0.022	0.318 ± 0.000

large-T limit. To derive Eq. (7), form the candidate residual $\varepsilon_i = s_i\,\tilde{g}_i$ for some scalar $s_i \in (0, 1]$ and require that the composed series have the same variance as the generator path:

$$\operatorname{Var}(g_i) \;=\; \operatorname{Var}(\alpha_i + \beta_i\, g_m + \varepsilon_i) \;=\; \operatorname{Var}(\beta_i\, g_m + \varepsilon_i) \;=\; \sigma^2_{\text{gen},i}, \tag{S18}$$

where α_i drops out because it is a constant offset. Expanding the variance of the sum:

$$\operatorname{Var}(\beta_i\, g_m + \varepsilon_i) \;=\; \beta_i^2\, \operatorname{Var}(g_m) \;+\; 2\beta_i\, \operatorname{Cov}(g_m,\, \varepsilon_i) \;+\; \operatorname{Var}(\varepsilon_i). \tag{S19}$$

Substituting $\varepsilon_i = s_i\,\tilde{g}_i$ yields $\operatorname{Var}(\varepsilon_i) = s_i^2\,\sigma^2_{\text{gen},i}$ and $\operatorname{Cov}(g_m,\, \varepsilon_i) = s_i\, \operatorname{Cov}(g_m,\, \tilde{g}_i)$, so under the independence assumption $\operatorname{Cov}(\tilde{g}_i,\, g_m) \approx 0$ the cross term vanishes and the variance condition reduces to $\beta_i^2\,\sigma_m^2 + s_i^2\,\sigma^2_{\text{gen},i} = \sigma^2_{\text{gen},i}$, which rearranges directly to Eq. (7).

To derive Eq. (10), write the population R^2 identity for Eq. (4):

$$R^2 \;=\; \frac{\beta_i^2\,\sigma_m^2}{\beta_i^2\,\sigma_m^2 + \sigma^2_{\varepsilon,i}}, \tag{S20}$$

and invert for the residual variance that matches the calibrated R^2; the result is the boxed Eq. (10). By construction the composed series g_i then satisfies the three preservation criteria in the large-T limit. OLS of g_i on g_m recovers $\hat{\alpha}_i \to \alpha_i$ and $\hat{\beta}_i \to \beta_i^{\mathrm{eff}}$, with $\beta_i^{\mathrm{eff}} = \beta_i$ unclipped and β_i^{eff} persisted alongside β_i when the clip triggers. Marginal variance matches $\sigma^2_{\mathrm{gen},i}$ on the variance-preserving branch, either exactly $\mathrm{Var}(g_i) = \beta_i^2 \sigma_m^2 + s_i^2 \sigma^2_{\mathrm{gen},i} = \sigma^2_{\mathrm{gen},i}$, or by the choice of β_i^{eff} in Eq. (9) under clipping; on the R^2-preserving branch the marginal variance instead matches $\beta_i^2 \sigma_m^2 / R^2_{i,\mathrm{real}}$, the value implied by the calibrated $(\beta_i, R^2_{i,\mathrm{real}})$ rather than by the generator marginal. Tail behavior on the variance-preserving branch follows from the floor: because at least $f\,\sigma^2_{\mathrm{gen},i}$ of the variance is reserved for ε_i, the tails of $\tilde{g}_i$ dominate whenever $(\beta_i^{\mathrm{eff}})^2 \sigma_m^2$ is small relative to those tails, while as β_i grows the marginal of g_i tightens toward Gaussian along the market-driven axis, mildly for low-β assets and noticeably at high β.